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From AI Capabilities to Sustainable Competitive Advantages: A Dynamic Capabilities Perspective in E-Commerce and Platform-Based Digital Business

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Abstract

Artificial intelligence (AI) has become a standard tool in e-commerce and platform-based business, yet its wide availability raises the question of how it can create an advantage that competitors cannot easily copy. Drawing on the resource-based view and dynamic capabilities theory, this thesis proposes that AI capability leads to sustainable competitive advantage through the dynamic capabilities of sensing, seizing and transforming, and evaluates to what extent published empirical evidence supports six hypotheses derived from this model.

A structured literature review was conducted. A search in OpenAlex identified 1,807 records, which were classified in two separate runs of a language model against explicit criteria (Cohen's kappa = 0.84 between runs). Of these, 120 studies were included, nine of them provisionally on the basis of their titles alone. Twenty of the 95 empirical studies could be appraised against nine quality criteria, and all results were classified with predefined decision rules.

The findings can be stated at three levels. First, the literature suggests that AI is associated with organisational outcomes, mostly through complementary capabilities such as agility, innovation capability and organisational learning; in some of the appraised studies the direct association between AI and competitive advantage became small or non-significant once these capabilities were included, whereas in others it remained. Second, the evidence partly supports the view that AI capability contributes to competitive advantage through such capabilities: no hypothesis is supported, and all six are at most partly supported, on very different bases. The direct relationships (H1, H2) rest on several studies of mostly unknown quality, each dimension-specific mediation (H3 to H5) on a single indirectly relevant study, and the higher-order mediation (H6) on studies of related capabilities rather than of dynamic capabilities themselves. Third, the proposed mediation through sensing, seizing and transforming and the persistence of any advantage over time remain unconfirmed: no study estimates a dimension-specific indirect effect on competitive advantage, most studies are cross-sectional, the quality of most could not be assessed, and only four empirical studies concern e-commerce or platform-based business, one of them on complementors that sell through platforms.

The thesis concludes that the existing evidence does not yet confirm the proposed mechanism for e-commerce and platform firms. Managers should therefore invest in the capabilities and governance surrounding AI rather than in the technology alone, and future research needs dimension-specific, longitudinal studies in this context.

Keywords: artificial intelligence capability, dynamic capabilities, sustainable competitive advantage, e-commerce, digital platforms, structured literature review

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AI Artificial Intelligence

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1 Introduction

1.1 Background to the Research and Research Justification

Artificial intelligence (AI) has moved from a specialist technology to a general-purpose resource that can augment or replace human tasks across industries, including retail, logistics and supply chains (Dwivedi et al., 2021). E-commerce is one of the fields in which this shift is most visible, and research on AI applications in online commerce has grown into a substantial body of work in its own right (Bawack et al., 2022). Platform-based businesses are particularly exposed to this development. Many of the most profitable firms own platforms that exhibit network effects, and advances in AI have added a further mechanism, data network effects, through which a platform becomes more valuable to each user the more it learns from the data it collects (Gregory et al., 2021). Machine learning can therefore strengthen the lock-in effects of a business model beyond the traditional direct and indirect network effects (Haftor et al., 2021).

This logic has an important implication for competition. If more data leads to more learning, and more learning attracts more users who generate more data, the resulting cycle favours firms that are already large (Haftor et al., 2021). AI would then reinforce existing positions rather than create new ones. The relationship is, however, not automatic. Gregory et al. (2021) argue that the link between a platform's AI capability and the value perceived by its users depends on platform legitimation, data stewardship and user-centric design. Whether AI produces an advantage therefore depends on how the platform organises and governs its use of AI, and not only on the volume of data it controls. This distinction between owning AI and organising around it is the starting point of this thesis.

Investment in AI does not translate automatically into superior performance. A systematic review of the AI business value literature concludes that organisations still struggle to adopt and leverage AI in their operations, and that the mechanisms through which AI creates value remain insufficiently understood (Enholm et al., 2022). The same review separates first-order effects of AI, such as process efficiency, from second-order effects at the level of the organisation, a distinction that is central to the question of how AI might lead to a lasting competitive advantage. Comparable difficulties have been reported when manufacturing firms attempt to scale AI capabilities across their business models and operations (Sjödin et al., 2021). AI can even have negative consequences: opaque, poorly governed AI-integrated analytics have been associated with suboptimal decisions, operational inefficiency and, ultimately, competitive disadvantage (Rana et

al., 2022). In that study, the opacity resulted from a lack of governance, poor data quality and insufficient training of key employees, which are organisational rather than technical shortcomings.

The positive evidence should also be read with care. An analysis of 500 AI deployment cases suggests that organisations achieve performance gains from AI capabilities only when they use them to reconfigure their processes (Wamba-Taguimdje et al., 2020). The finding is relevant because it points to reconfiguration, a core element of dynamic capabilities, as the condition under which AI creates value. The cases were, however, collected from the websites of AI solution providers, a source that is likely to over-represent successful deployments. The study therefore indicates what AI can achieve under favourable conditions rather than how often such outcomes occur.

The resource-based view (RBV) offers one explanation for these differences. From this perspective, value does not stem from AI technology as such but from an AI capability, that is, a firm-specific combination of AI-related tangible, human and intangible resources (Mikalef & Gupta, 2021). Empirical work supports a link between such a capability and firm outcomes, including organisational creativity and performance in the same study and, in a survey of 394 e-commerce firms, an indirect effect on performance through creativity, AI management and AI-driven decision making (Chen et al., 2022). In the latter study, the strength of these relationships also depended on contextual conditions such as innovation culture and environmental dynamism. AI may, moreover, change the sources of competitive advantage themselves. Using chess tournaments as a controlled setting, Krakowski et al. (2023) show that AI can make established capabilities obsolete while creating new sources of heterogeneity at the intersection of human and machine decision making. The advantage then lies neither in the human nor in the technology alone, but in how the two are combined.

Dynamic capabilities theory extends this reasoning to fast-changing environments. It shifts the attention from the resources a firm owns to its ability to sense changes, seize opportunities and transform its resource base. Case evidence from three large Norwegian organisations indicates that AI can act as a micro-foundation of such capabilities, helping firms to manage their operations in dynamic and uncertain environments (Mikalef et al., 2021). Conceptual work similarly argues that AI leads to competitive advantage only when it is “situated” in the firm through grounding, bounding and recasting activities, which make AI-driven capabilities firm-specific, cost-effective and suited to the opportunities in the strategic environment (Kemp, 2024). Taken together, these contributions suggest that dynamic capabilities may be the mechanism that connects an AI capability with an advantage that lasts.

This mechanism deserves particular attention in e-commerce and platform-based business. Competition in digital markets follows a logic that departs in important respects from the predictions of mainstream competitive theory, because the value of a platform depends both on its size and on its distinctive technological and market identity (Cennamo, 2021). Findings obtained in manufacturing or in traditional business-to-business settings can therefore not simply be transferred to these firms. A clearer understanding of how AI capabilities contribute to a lasting competitive advantage in this context is relevant for two groups. Managers need to decide whether to invest primarily in AI technology or in the organisational capabilities required to use it, and researchers need to explain why similar AI investments lead to different competitive outcomes.

1.2 Research Problem

As AI tools, cloud services and pre-trained models become widely accessible, the technology alone offers limited protection against imitation. Rival firms can acquire similar software and hire similar specialists, so an advantage is more likely to depend on how AI is embedded in the organisation than on access to the technology itself (Kemp, 2024; Krakowski et al., 2023). The central question is therefore not whether firms use AI, but through which organisational mechanisms an AI capability is converted into an advantage that persists over time.

The existing empirical evidence answers this question only in part. Several studies report a positive relationship between AI capability, or related constructs such as AI assimilation, and firm performance. They do so, however, through a wide range of intervening variables, including organisational creativity (Mikalef & Gupta, 2021; Chen et al., 2022), business-to-business marketing capabilities (Mikalef et al., 2023a), organisational and customer agility (Fosso Wamba, 2022) and business model innovation (Shao et al., 2026). Some of these mediators, such as agility or business model innovation, are conceptually related to sensing, seizing and transforming, but they are not identical with them. Only a few studies model the three dimensions of dynamic capabilities as distinct links between AI and competitive advantage. One recent study of 243 Spanish organisations and ten case studies relates AI integration to these dimensions and reports an overall indirect effect of AI-enabled dynamic capabilities on competitive advantage (Selgas-Cors & Thiébaut, 2026). Separate indirect effects for each dimension are not reported, so it remains open which of the three dimensions carries the effect.

A second difficulty concerns the constructs themselves. The literature does not use a single definition of AI capability. Mikalef & Gupta (2021) define it as a combination of AI-specific resources and treat it as a resource in the sense of the RBV. Chen et al.

(2022) model it as a higher-order construct formed by basic resources, proclivity and skills, and Shao et al. (2026) distinguish tangible, intangible and skill resources. Other studies examine AI assimilation (Fosso Wamba, 2022) or AI integration (Selgas-Cors & Thiébaut, 2026), while adoption studies focus on the earlier decision to introduce AI at all (Chatterjee et al., 2021). The conceptual status of AI capability is also contested: Shao and colleagues interpret it as a dynamic capability in its own right, whereas the RBV-based studies treat it as a resource that dynamic capabilities act upon. If AI capability is itself a dynamic capability, the proposed mediation through sensing, seizing and transforming partly overlaps with the independent variable. The thesis therefore needs to establish which construct each study actually measures before its findings can be compared.

The outcome variables are equally diverse. The studies cited above measure organisational creativity, firm performance, business-to-business marketing outcomes, competitive advantage and, in one case, competitive disadvantage (Mikalef & Gupta, 2021; Mikalef et al., 2023a; Shao et al., 2026; Rana et al., 2022). Performance and competitive advantage are related but distinct concepts: a firm can perform well in a given year without holding an advantage that rivals cannot imitate. Treating performance as a proxy for sustainable competitive advantage would therefore overstate what the evidence shows.

The evidence is also concentrated in contexts that differ from e-commerce and platform-based business. Much of it comes from manufacturing firms (Sjödin et al., 2021; Chatterjee et al., 2021), business-to-business marketing (Mikalef et al., 2021; Mikalef et al., 2023a), small and medium-sized enterprises in China (Shao et al., 2026), service firms in India (Rana et al., 2022) or cross-industry samples (Selgas-Cors & Thiébaut, 2026). Where e-commerce firms are studied directly, the outcome is general firm performance rather than a lasting advantage (Chen et al., 2022), while the work that explicitly addresses platforms is largely conceptual (Gregory et al., 2021; Cennamo, 2021). The research designs vary as well, ranging from surveys analysed with structural equation modelling to qualitative case studies and, in one case, a controlled setting outside business (Krakowski et al., 2023). The survey-based studies typically rely on cross-sectional data and on outcome measures such as performance, which cannot by themselves establish whether an advantage persists over time or in which direction the relationships run.

It therefore remains unclear to what extent the published evidence supports the proposition that AI capabilities lead to a sustainable competitive advantage in e-commerce and platform-based businesses through sensing, seizing and transforming capabilities. A further study that simply adds another cross-sectional survey would not resolve

this uncertainty, because the main obstacle is the fragmentation of the existing evidence. This thesis addresses the problem by systematically assessing the published empirical evidence for each of the proposed relationships, with explicit attention to how constructs were measured, in which context and with which research design.

1.3 Research Objectives and Research Questions

The overall aim of this thesis is to evaluate the extent to which published empirical studies support the proposition that AI capabilities contribute to a sustainable competitive advantage in e-commerce and platform-based businesses through dynamic capabilities. The evaluation takes account of differences in construct measurement, context and research design between the studies. To achieve this aim, the thesis pursues six objectives. The first two concern the direct relationships on which the model rests; the remaining four concern the proposed mediating mechanisms, which form the main contribution of the thesis:

1. To evaluate the evidence on the relationship between AI capabilities and sustainable competitive advantage.
2. To evaluate the evidence on the relationship between AI capabilities and dynamic capabilities.
3. To evaluate the evidence on the mediating role of sensing capability.
4. To evaluate the evidence on the mediating role of seizing capability.
5. To evaluate the evidence on the mediating role of transforming capability.
6. To evaluate the evidence on the mediating role of dynamic capabilities as a higher-order construct.

Objectives 3 to 5 and Objective 6 are not simply repetitions of the same question. The separate assessment of sensing, seizing and transforming shows where in the organisation AI creates value, which matters for managers who must decide where to direct their investment. The higher-order view in Objective 6 asks whether the three dimensions work as a combined capability whose effect is greater than that of each dimension alone. Because both perspectives may draw on the same studies, the evidence is recorded once per study and then assigned to the relevant objectives, so that no study is counted twice in support of the same claim. These objectives lead to four research questions:

RQ1: To what extent does published empirical evidence support a relationship between AI capabilities and sustainable competitive advantage?

RQ2: To what extent does published empirical evidence support a relationship between AI capabilities and dynamic capabilities?

RQ3: To what extent does published empirical evidence indicate that sensing, seizing and transforming capabilities each mediate the relationship between AI capabilities and sustainable competitive advantage?

RQ4: To what extent does published empirical evidence indicate that dynamic capabilities, as a higher-order construct, mediate this relationship?

Chapter 2 translates these questions into six hypotheses (H1–H6). Because the thesis is based on a structured review of existing studies, the hypotheses are assessed against the reported evidence rather than tested statistically with new data. A hypothesis can therefore be judged as supported, partly supported, inconclusive, contradicted or not yet examined by the available studies.

1.4 Scope and Limitations

The unit of analysis is the firm. The thesis focuses on e-commerce retailers and on firms that own and operate digital platforms. Complementors, such as independent sellers or app developers who offer their products through a platform, face different conditions regarding data access and governance. Studies of complementors are therefore treated as closely related evidence: they are counted as part of the wider e-commerce and platform context, but they are labelled as such and not taken as evidence about platform owners. Studies from other industries are included where they examine the proposed relationships, but their transferability to e-commerce and platform-based business is discussed explicitly rather than assumed.

In this thesis, a sustainable competitive advantage denotes an advantage over competitors that persists over time and resists imitation. The term does not refer to environmental sustainability. The distinction is necessary because parts of the related literature use the word in its ecological sense; Haftor et al. (2021), for example, examine how machine learning can increase profitability while reducing negative ecological impacts. Studies that use “sustainable” only in this environmental sense are not treated as evidence on the persistence of competitive advantage.

Several questions lie outside the scope of the thesis. It does not evaluate individual AI technologies or algorithms, does not examine the consumer perspective on AI in e-commerce and does not measure the capabilities of particular firms. It also does not make causal claims beyond those that the underlying studies can support. Where the evidence is observational, relationships are described as associations rather than effects.

Further limitations follow from the research design. The thesis does not collect primary data and can only be as strong as the published studies on which it is based. Many of these studies are cross-sectional, rely on managers' perceptions and examine organisations outside e-commerce, so their findings are transferable to the focal setting only to a limited extent. Finally, the review is restricted to studies that can be identified through the documented search procedure; the exact databases, time window and eligibility criteria are specified in Chapter 3.

1.5 Research Approach

The thesis follows a deductive approach. Hypotheses are derived from the resource-based view and dynamic capabilities theory in Chapter 2 and then compared with the findings of existing empirical studies. A structured literature review with a narrative synthesis is used for this purpose (Booth et al., 2025). This design suits the research questions for three reasons. First, the main uncertainty identified in Section 1.2 lies in the fragmentation of existing evidence, which a further single study could not resolve. Second, the relevant studies measure AI capability, dynamic capabilities and competitive advantage in different ways, which makes a statistical pooling of their results in a meta-analysis inappropriate. Third, original survey data from e-commerce and platform firms would be difficult to obtain within the scope of a master's thesis and would again be limited to a single, cross-sectional context.

The review proceeds in four steps. First, relevant studies are identified through a documented and reproducible database search. Second, explicit inclusion, exclusion and quality criteria are applied. The quality appraisal considers, among other aspects, whether a study separates its measures in time, how it addresses common method bias and whether indirect effects were actually estimated rather than inferred. Third, the evidence from each included study is recorded in an extraction matrix that distinguishes direct associations from reported indirect effects and documents the exact constructs measured. Fourth, the evidence for each hypothesis is classified as directly reported, indirectly relevant, contradictory or not available, taking the methodological quality of each study into account.

This approach has a clear limitation: it cannot produce new estimates of the proposed relationships. Its contribution lies instead in establishing, transparently and for each relationship separately, how much the current evidence can support, and in identifying where further empirical research is most needed.

1.6 Structure of the Thesis

The thesis consists of five chapters. Following this introduction, Chapter 2 reviews the theoretical foundations of the resource-based view and dynamic capabilities theory, defines the key constructs and examines the empirical literature on AI capability in e-commerce and platform-based business. It closes by identifying the research gap and developing the conceptual model and hypotheses. Chapter 3 then describes the research design, including the search strategy, the eligibility criteria, the quality appraisal and the synthesis rules.

Chapter 4 presents the results of the review. It reports the characteristics of the included studies, assesses the evidence for each hypothesis and discusses the findings in relation to the literature reviewed in Chapter 2. Chapter 5 summarises the findings, answers the research questions and discusses the theoretical and managerial implications, the limitations of the thesis and directions for future research.

2 Literature Review

This chapter reviews the literature on which the thesis builds. Section 2.1 introduces the resource-based view and dynamic capabilities theory. Section 2.2 defines the key constructs, and Section 2.3 examines the platform context and the empirical evidence on AI capability. Section 2.4 identifies the research gap and develops the conceptual model and hypotheses, and Section 2.5 summarises the chapter.

2.1 Theoretical Foundations

Two theoretical perspectives are combined in this thesis because each answers a different part of the research question. The resource-based view explains which resources can give a firm an advantage that competitors cannot easily copy, but it says little about how such resources are created and renewed when markets change. Dynamic capabilities theory addresses precisely this question: it explains how firms sense changes in their environment, seize the resulting opportunities and transform their resource base. For AI, both questions matter. AI can be treated as a resource whose value depends on how difficult it is to imitate, and at the same time as an enabler of the processes through which firms adapt. The following subsections introduce the two perspectives and their limitations.

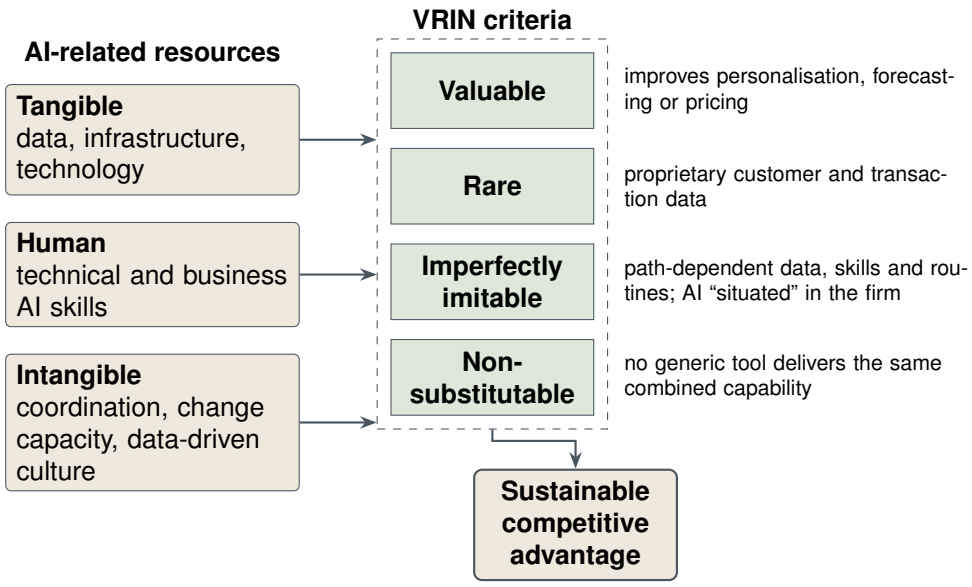
2.1.1 Resource-Based View

The resource-based view (RBV) explains differences in firm performance by differences in the resources that firms control. Barney (1991) builds this argument on two assumptions: strategic resources are distributed heterogeneously across firms, and these differences remain stable over time. A resource can then contribute to a sustained competitive advantage if it is valuable, rare, imperfectly imitable and non-substitutable, the four criteria commonly abbreviated as VRIN. Peteraf (1993) refines this logic by specifying four conditions that must all be met: superior resources, ex post limits to competition, imperfect resource mobility and ex ante limits to competition. Both contributions share a central point: an advantage is sustainable only if competitors cannot easily obtain or copy the resources on which it rests.

Applied to AI, the RBV suggests that AI technology alone is unlikely to meet these criteria. Algorithms, cloud infrastructure and pre-trained models are widely available, which makes them valuable but rarely rare or difficult to imitate. Research has therefore

shifted from AI technology to AI capability, understood as a firm-specific bundle of tangible, human and intangible AI-related resources (Mikalef & Gupta, 2021). Subsequent studies model this capability as a higher-order construct and identify AI infrastructure as a particularly important resource within it (Fosso Wamba et al., 2024). Figure 2.1 applies the VRIN criteria to such AI-related resources. It illustrates that the criteria are most likely to be met not by single technologies but by combinations of proprietary data, specialised skills and organisational routines, which are path-dependent and therefore hard to copy.

Figure 2.1: VRIN criteria applied to AI-related resources



Source: author's own work, adapted from Barney (1991) and Mikalef & Gupta (2021).

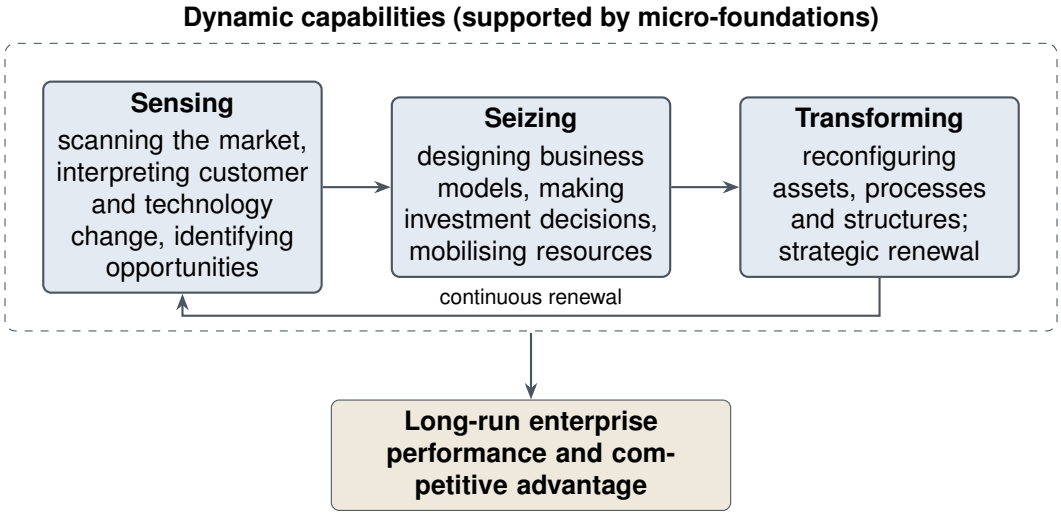
The RBV has, however, been criticised for treating resources as relatively static. Eisenhardt & Martin (2000) argue that the traditional RBV misidentifies the source of long-term advantage in dynamic markets and reaches a boundary condition in high-velocity markets, where advantages are temporary. This criticism is particularly relevant for AI. Krakowski et al. (2023) show that AI adoption can make established capabilities obsolete while new sources of heterogeneity emerge at the intersection of human and machine decision making. A resource that satisfies the VRIN criteria today may therefore lose its value quickly once AI changes the basis of competition. This limitation motivates the second theoretical perspective of the thesis.

2.1.2 Dynamic Capabilities Theory

Dynamic capabilities theory addresses how firms create and maintain advantages in environments of rapid technological change. Teece et al. (1997) locate competitive advantage in distinctive organisational processes, shaped by the firm's asset positions

and the paths it has followed, and argue that the ease of imitation and replication determines whether an advantage erodes. Teece (2007) later specifies dynamic capabilities as the capacities to sense opportunities and threats, to seize them and to reconfigure the firm's tangible and intangible assets. These capacities rest on micro-foundations, the distinct skills, processes, organisational structures and decision rules that are difficult to develop and deploy. Figure 2.2 summarises this framework.

Figure 2.2: Dynamic capabilities framework



Source: author's own work, adapted from Teece (2007).

The theory is not without tension. Eisenhardt & Martin (2000) describe dynamic capabilities as specific, identifiable processes, such as product development or strategic decision making, that show considerable commonalities across firms and resemble "best practice". If dynamic capabilities are more homogeneous and substitutable than often assumed, they cannot by themselves explain a lasting advantage; the advantage must lie in the resource configurations they produce. This argument is important for the present thesis, because it implies that sensing, seizing and transforming should be examined as mechanisms that convert AI capability into an advantage rather than as advantages in their own right.

Two extensions of the theory are directly relevant to digital and platform-based business. Helfat & Raubitschek (2018) argue that platform leaders need at least three types of dynamic capabilities, namely innovation capabilities, environmental scanning and sensing capabilities, and integrative capabilities for orchestrating the ecosystem, and that integrative capabilities are central to capturing value. Warner & Wäger (2019) show in a qualitative study of incumbent firms that dynamic capabilities for digital transformation are built through an ongoing process with nine micro-foundations, in which

agility is the core mechanism for renewing the business model, the collaborative approach and the culture. Both contributions suggest that in digital settings the transforming dimension is not a one-off adjustment but a continuous activity, which is reflected in the feedback loop in Figure 2.2.

2.2 Key Constructs

Several constructs in this field are used inconsistently, and the comparison of studies in Chapter 4 depends on distinguishing them precisely. Table 2.1 therefore sets out the working definitions used in this thesis. The following subsections explain and justify these definitions.

Table 2.1: Definitions of key constructs

Construct	Working definition in this thesis	Based on
AI adoption	The decision to introduce AI technologies into the organisation	Chatterjee et al. (2021)
AI integration / assimilation	The degree to which AI is embedded in business processes and routines	Fosso Wamba (2022); Selgas-Cors & Thiébaut (2026)
AI capability	A firm's ability to select, combine and deploy AI-specific tangible, human and intangible resources	Mikalef & Gupta (2021); Fosso Wamba et al. (2024)
Dynamic capabilities	Capacities to sense, seize and transform in response to environmental change, treated as a higher-order construct	Teece et al. (1997); Teece (2007)
Sensing	Identifying and interpreting opportunities and threats in markets and technologies	Teece (2007)
Seizing	Mobilising resources and making decisions to capture identified opportunities	Teece (2007)
Transforming	Reconfiguring assets, processes and the business model	Teece (2007); Warner & Wäger (2019)
Sustainable competitive advantage	An advantage over competitors that persists over time and resists imitation	Barney (1991); Peteraf (1993)

Source: author's own work.

2.2.1 AI Capability

AI itself is a moving target. Berente et al. (2021) describe it as the frontier of computational advancements that reference human intelligence, and argue that managing AI requires decisions about three interdependent facets: autonomy, learning and inscrutability. Huang & Rust (2021) distinguish mechanical AI for automating repetitive tasks, thinking AI for processing data to reach decisions, and feeling AI for analysing interactions and emotions. Raisch & Krakowski (2021) add that automation, in which

machines take over a task, and augmentation, in which humans and machines collaborate, cannot be neatly separated and create a paradoxical tension. Because the technology changes so quickly, a definition of AI capability that rests on specific technologies would soon be outdated. This is one reason why the literature defines AI capability in terms of organisational resources rather than tools.

A clear distinction between adoption, integration and capability follows from this (Table 2.1). Adoption studies explain why firms decide to introduce AI (Chatterjee et al., 2021), whereas integration or assimilation studies examine how far AI is embedded in operations (Fosso Wamba, 2022). AI capability goes further and describes the firm's ability to mobilise AI-related resources in combination. Mikalef & Gupta (2021) develop and validate a measurement instrument for this capability, and Chen et al. (2022) model it as a higher-order construct formed by basic resources, proclivity and skills. Qualitative research identifies the organisational capabilities that AI implementation requires, namely AI project planning, co-development, data management and AI model lifecycle management, which help firms to cope with the inscrutability and data dependency of AI (Weber et al., 2023).

These definitions are similar but not identical, and their differences matter for the present thesis. Shao et al. (2026) distinguish tangible, intangible and skill resources and interpret AI capability as a dynamic capability in its own right. This interpretation conflicts with the RBV-based view, in which AI capability is a resource bundle on which dynamic capabilities act. If AI capability were itself a dynamic capability, the proposed mediation through sensing, seizing and transforming would partly overlap with the independent variable. This thesis follows the resource-based definition of Mikalef & Gupta (2021), because it keeps the two constructs analytically separate and allows the mediating role of dynamic capabilities to be examined.

2.2.2 Sensing, Seizing and Transforming

Sensing, seizing and transforming describe different organisational activities, and AI can support each of them in different ways. Table 2.2 summarises the three dimensions, the role AI may play in each and illustrative applications in e-commerce and platform-based business.

For sensing, AI promises to widen and speed up the firm's view of its environment. In the marketing domain, thinking AI can support market analysis and feeling AI customer understanding (Huang & Rust, 2021). Case evidence identifies AI-specific microfoundations of dynamic capabilities that help organisations to operate in dynamic and uncertain environments (Mikalef et al., 2021), and survey evidence from 317 small

Table 2.2: Dimensions of dynamic capabilities and the role of AI in e-commerce

Dimension	Definition	Possible role of AI and illustrative e-commerce applications	Supporting literature
Sensing	Identifying and interpreting opportunities and threats	Thinking AI for market analysis and feeling AI for customer understanding; e.g. demand forecasting, analysis of reviews and search behaviour, monitoring competitor prices	Teece (2007); Huang & Rust (2021); Mikalef et al. (2021)
Seizing	Mobilising resources and taking decisions to capture opportunities	Faster and better-informed decisions; e.g. dynamic pricing, personalised recommendations, rapid assortment and promotion decisions	Teece (2007); Neiroukh et al. (2025); Chen et al. (2022)
Transforming	Reconfiguring assets, processes and the business model	Process reconfiguration and business model innovation; e.g. automated fulfilment and service processes, new data-driven services	Teece (2007); Wamba-Taguimdje et al. (2020); Sjödin et al. (2021)

Source: author's own work, based on the literature listed in the table; the e-commerce applications are illustrative.

and medium-sized enterprises (SMEs) in London shows that AI applications for targeting consumers online and forecasting cash flows were associated with reduced business risks during the COVID-19 pandemic (Drydakis, 2022). In e-commerce, where customer behaviour leaves extensive digital traces, the potential contribution of AI to sensing is particularly large.

For seizing, the main contribution of AI lies in decision making. A survey of 230 organisations finds that AI capability is positively related to both the speed and the quality of decisions, and that decision speed partly mediates the relationship with organisational performance (Neiroukh et al., 2025). In e-commerce firms, AI-driven decision making has been identified as one of the paths through which AI capability affects firm performance (Chen et al., 2022). A strategic review similarly lists decision support, automation and new products and services among the main sources of value creation from AI (Borges et al., 2021). Faster decisions are, however, not necessarily better decisions: Rana et al. (2022) show that opaque AI-based analytics can lead to suboptimal decisions and, ultimately, to competitive disadvantage.

For transforming, AI matters because value appears to depend on reconfiguration. Organisations achieve performance gains from AI only when they use it to reconfigure their processes (Wamba-Taguimdje et al., 2020), and manufacturers scale AI by innovating their business models through customer co-creation, data-driven delivery and ecosystem integration (Sjödin et al., 2021). Business model innovation also mediates the relationship between AI capability and competitive advantage in Chinese SMEs (Shao et al., 2026). These findings suggest that transforming may be the dimension

through which AI has its most lasting effect. They also illustrate a measurement problem: business model innovation and process reconfiguration are related to transforming, but they are not the same construct, and studies rarely state how their mediators map onto the three dimensions.

2.2.3 Sustainable Competitive Advantage

The dependent variable requires particular care. In the RBV tradition, a competitive advantage is sustained when it persists despite competitors' attempts to imitate it, not merely when it lasts for a certain period (Barney, 1991). Peteraf (1993) add that limits to competition must exist both before and after the firm obtains its superior position. Teece (2007) links dynamic capabilities to superior long-run enterprise performance and refers to it as sustainable performance. All three contributions describe the persistence of an advantage relative to competitors, which is the meaning adopted in this thesis.

Two other uses of the term must be distinguished from this meaning. First, "sustainable" increasingly appears in an environmental sense; Fosso Wamba et al. (2024), for instance, examine AI capability and firm performance from a sustainable development perspective. Second, many studies measure firm performance or productivity and interpret it as competitive advantage. This substitution is problematic. Brynjolfsson et al. (2021) show that general-purpose technologies such as AI require large intangible complementary investments, which can cause productivity to be underestimated in early years and overestimated later. A performance measure taken at a single point in time may therefore say little about whether an AI-based advantage persists. For this reason, the analysis in Chapter 4 records separately whether a study measures performance, competitive advantage or the persistence of that advantage.

2.3 AI in E-Commerce and Platform Businesses

The general theories introduced above do not specify how AI creates value in a particular industry. E-commerce and platform-based businesses are a distinctive setting for this question, because their operations generate large amounts of data on customer behaviour and their core processes, such as search, recommendation and pricing, are already digital. AI can therefore be embedded more directly into everyday operations than in many traditional industries, while competitors can often acquire similar tools just as quickly. This combination of high potential and low barriers to imitation makes the context particularly suitable for examining whether AI leads to a lasting advantage. The following subsections first describe the characteristics of platform-based business and then review the empirical evidence on AI capability.

2.3.1 Characteristics of Platform-Based Business

Platform-based businesses differ from traditional firms in ways that affect how AI can create an advantage. Platforms facilitate transactions and govern interactions between two or more distinct user groups who are connected through indirect network effects. A systematic review of 333 articles on platform competition identifies four recurring themes: network effects that create winner-takes-all dynamics, the interaction between platform strategy and corporate decisions such as vertical integration, the role of heterogeneity among platforms and users, and the orchestration of value creation and capture in the wider ecosystem (Rietveld & Schilling, 2021). The value of a platform depends not only on its size but also on its distinctive technological and market identity, so that competition in digital markets departs from the predictions of mainstream competitive theory (Cennamo, 2021). Platforms are, moreover, embedded in ecosystems of complementors whose activities depend on one another and which cannot be fully controlled through hierarchy (Jacobides et al., 2018).

AI adds a further source of value to this setting. With data network effects, a platform becomes more valuable to each user the more it learns from the data it collects (Gregory et al., 2021), and machine learning can extend the lock-in effects of a business model beyond direct and indirect network effects (Haftor et al., 2021). This mechanism would favour large incumbents and could make AI-based advantages self-reinforcing. The effect is, however, conditional. The same authors propose that platform legitimation, data stewardship and user-centric design moderate the link between AI capability and user value. A two-wave survey of 220 users of tourism platforms confirms that AI capability can raise perceived user value through a user-centric design experience, but can also lower it by increasing perceived privacy risk; data stewardship strengthened both effects (Yu et al., 2026). AI capability on platforms can therefore work in two directions at once.

These characteristics shape what sensing, seizing and transforming mean for platform firms. Helfat & Raubitschek (2018) emphasise integrative capabilities for orchestrating the ecosystem, which go beyond the internal reconfiguration on which most dynamic capabilities research focuses. Platform owners also differ from e-commerce retailers, which sell products themselves, and from complementors, which depend on the platform's data and rules. Evidence from one of these groups can therefore not simply be transferred to another, which is why this thesis focuses on retailers and platform owners and treats evidence from complementors as closely related but distinct (Section 1.4).

2.3.2 Empirical Evidence on AI Capability

Empirical research on AI capability has grown rapidly since 2020, and Table 2.3 summarises the studies that are most relevant to the proposed relationships. Four strands can be distinguished. The first strand links AI capability to firm performance. AI capability has been associated with organisational creativity and performance (Mikalef & Gupta, 2021), with performance in e-commerce firms through creativity, AI management and AI-driven decision making (Chen et al., 2022), and with performance through decision speed (Neiroukh et al., 2025). Further studies relate AI capability to performance through a data-driven culture (Fosso Wamba et al., 2024), link AI capabilities with open innovation and business performance in multinational business-to-business firms (Sahoo et al., 2024), and show that coupling AI capability with strategic agility supports product and service creativity in Chinese organisations (Ameen et al., 2024). Not all effects are positive: in 168 European municipalities, AI capabilities improved process automation and cognitive insight, but cognitive engagement had a negative effect on organisational performance (Mikalef et al., 2023b).

The second strand examines dynamic capabilities as a mediator. Research on big data analytics provides an important precedent: the effect of big data analytics capability on competitive performance in 202 Norwegian firms was not direct but fully mediated by dynamic capabilities (Mikalef et al., 2020). Studies of AI report similar patterns. AI capability has been found to be positively associated with dynamic capabilities, which are in turn associated with organisational creativity and performance (Almheiri et al., 2024; Almheiri et al., 2025). Dynamic capabilities also mediate the relationship between AI and supply chain resilience in SMEs (Le & Behl, 2024), and in a survey of 487 SME managers and owners, AI applications and dynamic capabilities both acted as mediators between environmental turbulence and firm performance (Al Dhaheri et al., 2024). Using archival data from Chinese listed firms and 511 paired questionnaires, Gao et al. (2025) show that AI use affects innovation capability through digital adaptability and market perception. Further work examines the role of dynamic capabilities in the relationship between AI and operational performance (Wong & Ngai, 2025).

The third strand addresses competitive advantage directly. In 546 Chinese SMEs, AI capability was positively associated with competitive advantage, with business model innovation acting as a mediator (Shao et al., 2026), and in a cross-industry study of Spanish organisations, AI-enabled dynamic capabilities carried an overall indirect effect on competitive advantage (Selgas-Cors & Thiébaut, 2026). Other studies relate marketing analytics capability and AI adoption to competitive advantage in manufacturing (Hossain et al., 2022) and examine how generative AI capabilities contribute to competitive advantage through service innovation under environmental dynamism (Li

et al., 2025). In manufacturing, AI capabilities have also been linked to servitization, with absorptive capacity strengthening the relationship (Abou-Foul et al., 2023). The fourth strand is conceptual. Kemp (2024) argues that AI must be situated in the firm to produce an advantage, and Moser et al. (2024) respond that situatedness must be taken even more seriously when theorising about AI-based competitive advantage.

Table 2.3: Comparison of key prior studies

Study	Context and sample	Design	AI construct	Mediator(s)	Outcome
Mikalef et al. (2020)	202 Norwegian firms (CIOs, IT managers)	Survey, PLS-SEM	Big data analytics capability	Dynamic capabilities; marketing and technological capabilities	Competitive performance
Chen et al. (2022)	394 e-commerce firms	Survey, PLS-SEM	AI capability (higher-order)	Creativity; AI management; AI-driven decisions	Firm performance
Drydakis (2022)	317 SMEs, London	Survey with follow-up	AI applications	–	Business risks (COVID-19)
Mikalef et al. (2023a)	155 European firms	Survey, PLS-SEM	AI competencies	B2B marketing capabilities	Organisational performance
Mikalef et al. (2023b)	168 municipalities (NO, DE, FI)	Survey, SEM	AI capabilities	Automation; insight; engagement	Organisational performance
Abou-Foul et al. (2023)	185 manufacturers (US, EU)	Survey, SEM and fsQCA	AI capabilities	Moderator: absorptive capacity	Servitization
Almheiri et al. (2024); Almheiri et al. (2025)	344 public managers, UAE (same data set)	Survey, scale development	AI capability	Dynamic capabilities; creativity	Organisational performance
Ameen et al. (2024)	Organisations in China	Survey of managers	AI capability coupled with strategic agility	–	Product and service creativity
Al Dhaheri et al. (2024)	487 SME managers and owners	Survey	AI applications (as mediator)	AI applications and dynamic capabilities, between turbulence and outcomes	Innovation; firm performance
Neiroukh et al. (2025)	230 organisations	Survey, PLS-SEM	AI capability	Decision speed and quality	Organisational performance

Table 2.3 continued

Study	Context and sample	Design	AI construct	Mediator(s)	Outcome
Gao et al. (2025)	Chinese listed firms; 511 paired questionnaires	Archival data and survey	AI use	Digital adaptability; market perception	Innovation capability
Selgas-Cors & Thiébaut (2026)	243 Spanish organisations; 10 cases	Survey and case studies	AI integration	Sensing, seizing, reconfiguring (overall effect)	Competitive advantage
Shao et al. (2026)	546 Chinese SMEs	Survey, SEM	AI capability	Business model innovation	Competitive advantage
Yu et al. (2026)	220 tourism platform users	Two-wave survey	Platform AI capability	User-centric design; privacy risk	Perceived user value

Source: author's own work, based on the cited studies.

Read together, the studies in Table 2.3 support a cautious conclusion. AI capability is consistently associated with positive outcomes, and several studies find that dynamic capabilities or related constructs carry at least part of this effect. The evidence has, however, four weaknesses that limit what can be inferred. First, almost all studies rely on cross-sectional surveys of managers; exceptions are the follow-up design of Drydakis (2022), the two-wave survey of Yu et al. (2026) and the combination of archival and survey data in Gao et al. (2025). Second, the mediators differ from study to study, and only Selgas-Cors & Thiébaut (2026) model sensing, seizing and reconfiguring as separate dimensions. Third, the samples come mainly from SMEs, manufacturing, public organisations and cross-industry settings; only Chen et al. (2022) study e-commerce firms, and only the two-wave survey mentioned above studies a platform, from the user's perspective. Fourth, some evidence is less independent than it appears: the two studies by Almheiri and colleagues use the same sample of 344 public managers (Almheiri et al., 2024; Almheiri et al., 2025) and must be treated as one source of evidence rather than two.

2.4 Research Gap and Conceptual Model

A research gap exists where the available literature does not answer a question that theory considers important. Identifying such a gap requires comparing what theory predicts with what empirical studies have actually examined. In this thesis, the resource-based view and dynamic capabilities theory predict a specific mechanism, in which AI capability leads to competitive advantage through sensing, seizing and transforming, while the empirical review above shows how far this mechanism has been studied. The

first subsection sets out where the two diverge, and the second translates the theoretical argument into a conceptual model and hypotheses that can be assessed against the evidence.

2.4.1 Research Gap

The review points to four related gaps. The first is a gap in context. The mediating role of dynamic capabilities between AI and firm outcomes has been examined in public organisations (Almheiri et al., 2025), SMEs (Al Dhaheri et al., 2024; Le & Behl, 2024), manufacturing (Abou-Foul et al., 2023) and cross-industry samples (Selgas-Cors & Thiébaut, 2026), but not in e-commerce retailers or platform owners. Given the specific competitive logic of platforms (Cennamo, 2021; Rietveld & Schilling, 2021), this is a substantial omission.

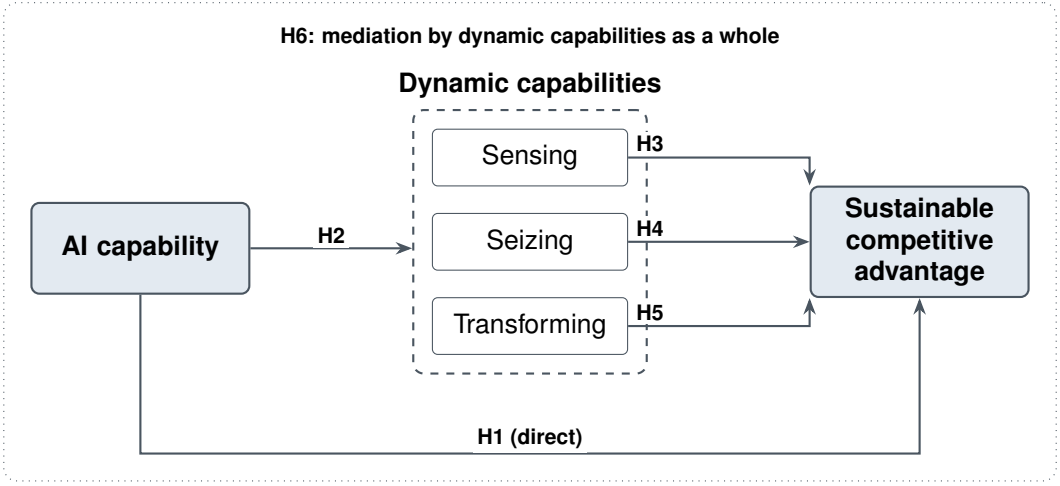
The second gap concerns the level of analysis within dynamic capabilities. Most studies treat dynamic capabilities as a single construct, and the one study that distinguishes sensing, seizing and reconfiguring reports only an overall indirect effect (Selgas-Cors & Thiébaut, 2026). It therefore remains unknown which dimension carries the effect of AI. The third gap concerns the outcome. Most studies measure performance, creativity or innovation rather than competitive advantage, and none measures whether an advantage persists over time. As Brynjolfsson et al. (2021) show, short-term performance can be a misleading indicator for technologies that require large complementary investments. The fourth gap is methodological: the predominance of cross-sectional, single-respondent surveys, together with the reuse of data sets, means that the existing evidence is thinner than the number of publications suggests.

These gaps justify the approach of this thesis. Rather than adding a further cross-sectional study, the thesis assesses systematically, relationship by relationship, how far the existing evidence supports the proposed mechanism, while accounting for differences in construct, context and design.

2.4.2 Conceptual Model and Hypotheses

Figure 2.3 presents the conceptual model of the thesis. AI capability is expected to relate to sustainable competitive advantage both directly and indirectly through dynamic capabilities. The indirect path is specified in two ways: through each of the three first-order dimensions of sensing, seizing and transforming, and through dynamic capabilities as a higher-order construct. The model applies to e-commerce retailers and platform owners.

Figure 2.3: Conceptual model of the thesis



Context: e-commerce retailers and platform owners

Source: author's own work.

The direct relationship (H1) follows from the RBV. If AI capability is a firm-specific combination of resources that is difficult to imitate, it should be associated with an advantage over competitors (Mikalef & Gupta, 2021; Kemp, 2024). Empirical support exists for related outcomes, such as competitive advantage in SMEs (Shao et al., 2026) and performance in e-commerce firms (Chen et al., 2022). The evidence from big data analytics, where the direct effect disappeared once dynamic capabilities were considered (Mikalef et al., 2020), suggests, however, that the direct relationship may be weak.

H1: AI capability is positively associated with sustainable competitive advantage.

The relationship between AI capability and dynamic capabilities (H2) follows from the micro-foundations argument of Teece (2007). AI can provide skills, processes and decision rules that support sensing, seizing and transforming, as case evidence (Mikalef et al., 2021) and survey evidence (Almheiri et al., 2025; Le & Behl, 2024) indicate. The relationship is expected for each dimension, but it is stated for dynamic capabilities as a whole, because the separate paths are covered by H3 to H5.

H2: AI capability is positively associated with dynamic capabilities.

The three dimension-specific hypotheses (H3 to H5) state that each dimension transmits part of the effect of AI capability. Sensing should matter because AI widens and accelerates the firm's view of markets and customers (Huang & Rust, 2021; Drydakis, 2022). Seizing should matter because AI improves the speed and quality of decisions (Neiroukh et al., 2025; Chen et al., 2022). Transforming should matter because AI

creates value mainly when processes and business models are reconfigured (Wamba-Taguimdje et al., 2020; Sjödin et al., 2021; Shao et al., 2026).

H3: Sensing capability mediates the relationship between AI capability and sustainable competitive advantage.

H4: Seizing capability mediates the relationship between AI capability and sustainable competitive advantage.

H5: Transforming capability mediates the relationship between AI capability and sustainable competitive advantage.

Finally, H6 treats sensing, seizing and transforming as a higher-order construct. This specification reflects the view that the three dimensions work as an integrated system whose effect may exceed that of each dimension alone (Teece, 2007), and it corresponds to studies that model dynamic capabilities as an overall construct (Mikalef et al., 2020; Selgas-Cors & Thiébaut, 2026). H6 is therefore not a repetition of H3 to H5 but an alternative specification of the same mechanism. To avoid counting the same evidence twice, a study that reports only an overall indirect effect is assigned to H6, and a study that reports dimension-specific effects is assigned to H3 to H5.

H6: Dynamic capabilities, as a higher-order construct, mediate the relationship between AI capability and sustainable competitive advantage.

Table 2.4 summarises the six hypotheses, their theoretical basis and the main prior evidence on which they draw.

2.5 Chapter Summary

This chapter has combined the resource-based view and dynamic capabilities theory to explain how AI capability may lead to a sustainable competitive advantage. The RBV explains why AI technology alone is unlikely to provide such an advantage, while dynamic capabilities theory explains how firms can turn AI-related resources into an advantage that survives changing conditions. The chapter has defined the key constructs, distinguishing AI adoption, integration and capability, and has separated sustainable competitive advantage from both environmental sustainability and short-term performance.

The review of the empirical literature shows that AI capability is consistently associated with positive outcomes and that dynamic capabilities appear to carry part of this effect. The evidence is, however, concentrated outside e-commerce and platform-based

Table 2.4: Summary of hypotheses

No.	Hypothesis	Theoretical basis	Main prior evidence
H1	AI capability is positively associated with sustainable competitive advantage	RBV	Shao et al. (2026); Chen et al. (2022)
H2	AI capability is positively associated with dynamic capabilities	Dynamic capabilities (micro-foundations)	Almheiri et al. (2025); Le & Behl (2024); Mikalef et al. (2021)
H3	Sensing mediates the relationship between AI capability and sustainable competitive advantage	Dynamic capabilities	Selgas-Cors & Thiébaut (2026) (path to sensing only)
H4	Seizing mediates the relationship	Dynamic capabilities	Selgas-Cors & Thiébaut (2026) (path to seizing only); Neiroukh et al. (2025)
H5	Transforming mediates the relationship	Dynamic capabilities	Selgas-Cors & Thiébaut (2026) (path to reconfiguring only); Shao et al. (2026)
H6	Dynamic capabilities, as a higher-order construct, mediate the relationship	RBV and dynamic capabilities	Selgas-Cors & Thiébaut (2026); Mikalef et al. (2020)

Source: author's own work.

business, rarely distinguishes the three dimensions of dynamic capabilities, seldom measures competitive advantage itself and relies mainly on cross-sectional data. On this basis, the chapter has developed a conceptual model with six hypotheses. The following chapter describes the method used to assess how far the published evidence supports each of them.

3 Methodology

This chapter describes how the research questions are answered. Section 3.1 sets out the research philosophy and the review design, and Section 3.2 explains how studies are searched for and selected. Section 3.3 describes the quality appraisal, the data extraction and the rules used to synthesise the evidence. Sections 3.4 and 3.5 discuss research quality, ethics and the limitations of the method, and Section 3.6 summarises the chapter.

3.1 Research Philosophy and Design

Methodological choices are not merely technical decisions. They follow from assumptions about the nature of the phenomena under study and about how knowledge of them can be obtained, and a research design is convincing only if these assumptions, the design and the methods are consistent with each other. A study that aims to assess hypotheses derived from theory, as this thesis does, requires a design in which the hypotheses are fixed in advance and the evidence is evaluated against explicit standards. The following subsections first set out the philosophical position of the thesis and then explain the review design that follows from it.

3.1.1 Research Paradigm

The research questions of this thesis ask to what extent published evidence supports a set of relationships that were specified in advance. This places the thesis within the positivist tradition. Positivism is aligned with the hypothetico-deductive model of science, in which hypotheses are formulated a priori, variables are operationalised and empirical findings are used to identify explanatory associations or causal relationships (Park et al., 2020). The reasoning of the thesis is accordingly deductive: it moves from the theory developed in Chapter 2 to the evaluation of evidence.

The positivist position is, however, qualified in one important respect. Ontologically, the thesis assumes that organisational capabilities and competitive positions exist independently of the researcher. Epistemologically, it assumes that these phenomena can only be known imperfectly, because the studies on which the thesis relies observe them through survey measures, managerial perceptions and particular samples. The thesis therefore adopts a post-positivist stance: empirical findings are treated as fallible and probabilistic, and the conclusions are expressed as degrees of support rather than as proof. With regard to axiology, the third assumption that distinguishes research

paradigms (Park et al., 2020), the influence of the researcher is limited by fixing the selection criteria and decision rules before the synthesis begins and by documenting every decision (Section 3.4).

An interpretivist approach, for example interviews exploring how managers make sense of AI, would be suitable for developing new explanations, but not for assessing how far the existing evidence supports relationships that are already specified. The evidence assessed in this thesis is mainly quantitative. Qualitative case studies are nevertheless retained where they illuminate the proposed mechanisms, and the synthesis itself is narrative rather than statistical, as Section 3.3.2 explains.

3.1.2 Structured Literature Review Design

Review research uses prior research as its data source in order to develop knowledge contributions (Kunisch et al., 2023), and a literature review can constitute an independent study with its own methods and contribution (Kraus et al., 2022). Reviews differ in how they search for, appraise, synthesise and analyse the literature. Grant & Booth (2009) compare fourteen review types along these four dimensions, which they summarise as the SALSA framework, and Snyder (2019) distinguishes systematic, semi-systematic and integrative approaches for business research. Traditional narrative reviews, by contrast, are often criticised for lacking thoroughness and for being open to researcher bias (Tranfield et al., 2003), a criticism that Snyder also shares.

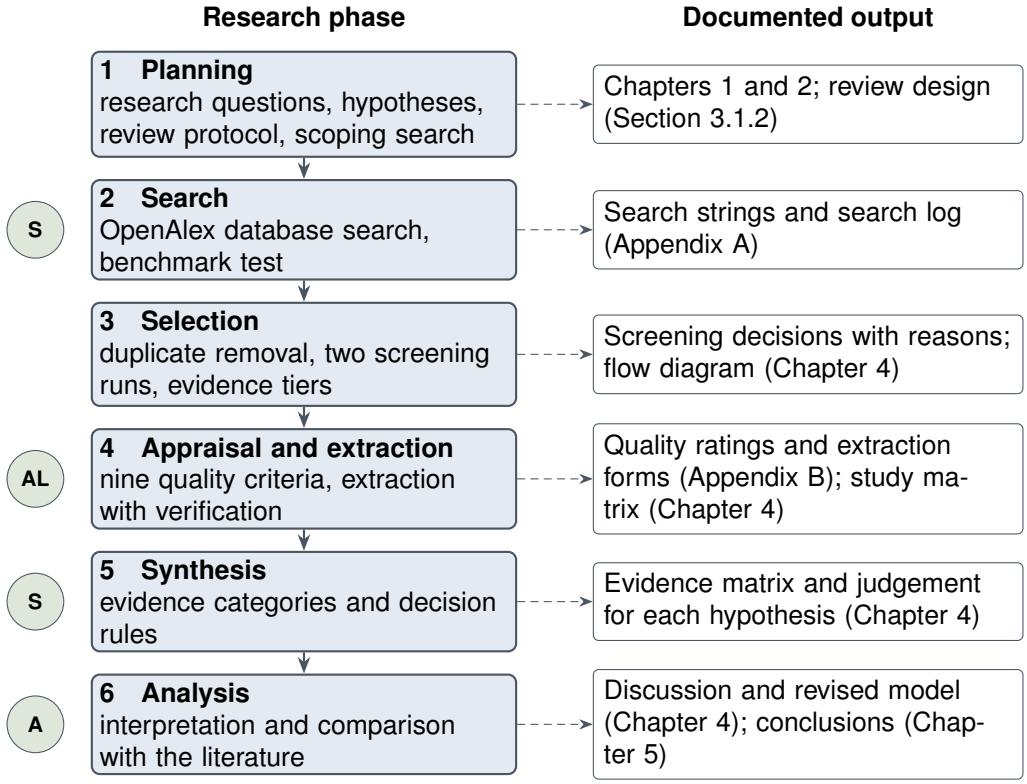
This thesis uses a structured literature review. It adopts the explicit search, selection and appraisal procedures of systematic reviews (Tranfield et al., 2003; Paul et al., 2021), but synthesises the evidence narratively rather than statistically. Systematicity is understood as a matter of degree: Simsek et al. (2023) describe it as an orientation towards explicit methods guided by the principles of transparency, coverage, saturation, connectedness, universalism and coherence, and warn against applying procedures without mindfulness. The review gives priority to transparency and coherence, because its purpose is evaluative. Kunisch et al. (2023) argue that the methods of a review must be aligned with its purpose, and assessing hypotheses requires explicit appraisal criteria and decision rules rather than a descriptive mapping of a field.

Three alternative designs were considered. A survey of e-commerce and platform firms, analysed with structural equation modelling, would produce new estimates. Access to such firms is, however, limited within a master's thesis, and a single cross-sectional survey would face the same problems of common method bias and uncertain temporal ordering that already limit the existing evidence (Podsakoff et al., 2024; Rohrer et al., 2022). A meta-analysis would require a sufficient number of studies with

comparable constructs and effect estimates, which Chapter 2 showed are not available; reviews in this situation can instead use structured methods of synthesis without meta-analysis (Campbell et al., 2020). A bibliometric analysis would map publication and citation patterns, but it could not evaluate how strongly the evidence supports specific relationships. The structured review is therefore the design that best matches the research questions.

Figure 3.1 summarises the research process. The process follows the search, appraisal, synthesis and analysis steps of the SALSA framework (Grant & Booth, 2009), adds planning and selection as separate phases and assigns a documented output to every phase. These outputs allow another researcher to retrace and repeat each step of the review.

Figure 3.1: Research process of the structured literature review



Source: author's own work; the circles mark the search (S), appraisal (AL), synthesis (S) and analysis (A) steps of the SALSA framework (Grant & Booth, 2009).

The unit of analysis of the thesis is the firm (Section 1.4). The unit of observation is the reported result, that is, each relationship between an AI construct, dynamic capabilities and a firm-level outcome that an included study reports. Because studies cannot be sampled at random from a known population, the selection of studies is criterion-based and therefore non-probabilistic. Its quality depends on how transparently the critical choices in the selection process are made and documented (Hiebl, 2023).

3.2 Search and Selection

In a literature review, the search and the selection of studies take the place of sampling in empirical research. They determine which evidence enters the analysis and therefore which conclusions are possible. Two decisions are central: where to search for studies, and which of the studies found are eligible. Both decisions have to be documented in enough detail to allow another researcher to repeat them, because a review whose evidence base cannot be reconstructed cannot be checked. The following subsections describe the search strategy and the criteria used to select studies.

3.2.1 Search Strategy

The choice of search systems determines which studies a review can find. Gusenbauer & Haddaway (2020) compared 28 academic search systems and found substantial differences in their precision, recall and reproducibility. Only half of the systems could be recommended for evidence syntheses without substantial caveats, and Google Scholar proved inappropriate as a principal search system. The search was therefore carried out in OpenAlex, an open bibliographic database that supports Boolean queries on titles and abstracts through a documented interface. Because its data are freely available, a search in OpenAlex can be reproduced by others without licensing barriers, and its reference coverage has been found comparable to that of Web of Science and Scopus (Culbert et al., 2025). The same comparison shows, however, that OpenAlex contains fewer abstracts, a limitation discussed in Section 3.5.

The search string combines three blocks of terms with the Boolean operator AND. The first block covers AI (*artificial intelligence*, *machine learning*, *AI*), the second covers capabilities (*capability*, *capabilities*, *sensing*, *seizing*, *reconfiguring*, *reconfiguration*) and the third covers outcomes (*competitive advantage*, *firm performance*, *organisational performance*, *competitive performance*). Terms within each block are combined with OR and searched in titles and abstracts; OpenAlex applies word stemming, so that grammatical variants are found without wildcards. Context terms such as *e-commerce* or *platform* are deliberately not used as a filter, because Chapter 2 showed that most of the relevant evidence comes from other settings; the context of each study is instead recorded during data extraction.

The search is limited to journal articles in English published from January 2015 onwards. This start date reflects that research on the strategic use of AI has grown mainly over the last decade, driven by advances in machine learning (Borges et al., 2021). No subject-area filter is applied. A test showed that the automatic subject classification of OpenAlex assigns at least one central study, Chen et al. (2022), to computer science, so that a filter for business and management would have excluded it. Before the main

search, the string was tested against known relevant studies, and it retrieved all three of them: Mikalef & Gupta (2021), the study by Chen and colleagues mentioned above, and Shao et al. (2026). The exact query, the search date and the number of records are documented in Appendix A. Duplicates are removed by OpenAlex identifier, DOI and normalised title.

Backward and forward citation searching of the included studies was planned as a supplementary step. It identified 678 further records that were not part of the database search, but these records could not be screened within the time available for this thesis. They are therefore not part of the evidence base, and this omission is treated as a limitation (Section 3.5).

3.2.2 Inclusion and Exclusion Criteria

Studies are selected against the criteria in Table 3.1. The criteria define which publication types, languages, periods, study types, constructs, units of analysis and data qualify a study for inclusion, and every exclusion is linked to a reason code (E1–E7).

Table 3.1: Inclusion and exclusion criteria

Criterion	Inclusion	Exclusion (reason code)
Publication type	Peer-reviewed journal articles, including articles published online first	Conference papers, working papers, preprints, theses, books and practitioner reports (E1)
Language	English	Other languages (E2)
Period	January 2015 to the date of the search	Published before 2015 (E3)
Study type	Empirical studies (quantitative, qualitative or mixed methods); conceptual studies are retained as Tier 3 context	Literature reviews, bibliometric studies, editorials, commentaries and book reviews (E4)
Constructs	An organisational AI construct (AI capability, integration, assimilation, use or competencies) and at least one of: dynamic capabilities or one of their dimensions, competitive advantage	AI studied only as a technology or adoption decision, or neither dynamic capabilities nor competitive advantage examined, including studies of firm performance only (E5)
Unit of analysis	Firm, organisation or business unit	Individuals, industries or countries; relevant studies may be retained as Tier 3 context (E6)
Data	Published findings with a reported analysis	Data sets without a peer-reviewed analysis (E7)
Context	Any industry and country; e-commerce and platform-based businesses are flagged	–

Source: author's own work.

Two of these decisions require explanation. First, the evidence base is restricted to peer-reviewed journal articles. Grey literature can increase the relevance of reviews in management research (Adams et al., 2017), but conference papers, working papers and deposited data sets have not passed peer review, and their results cannot

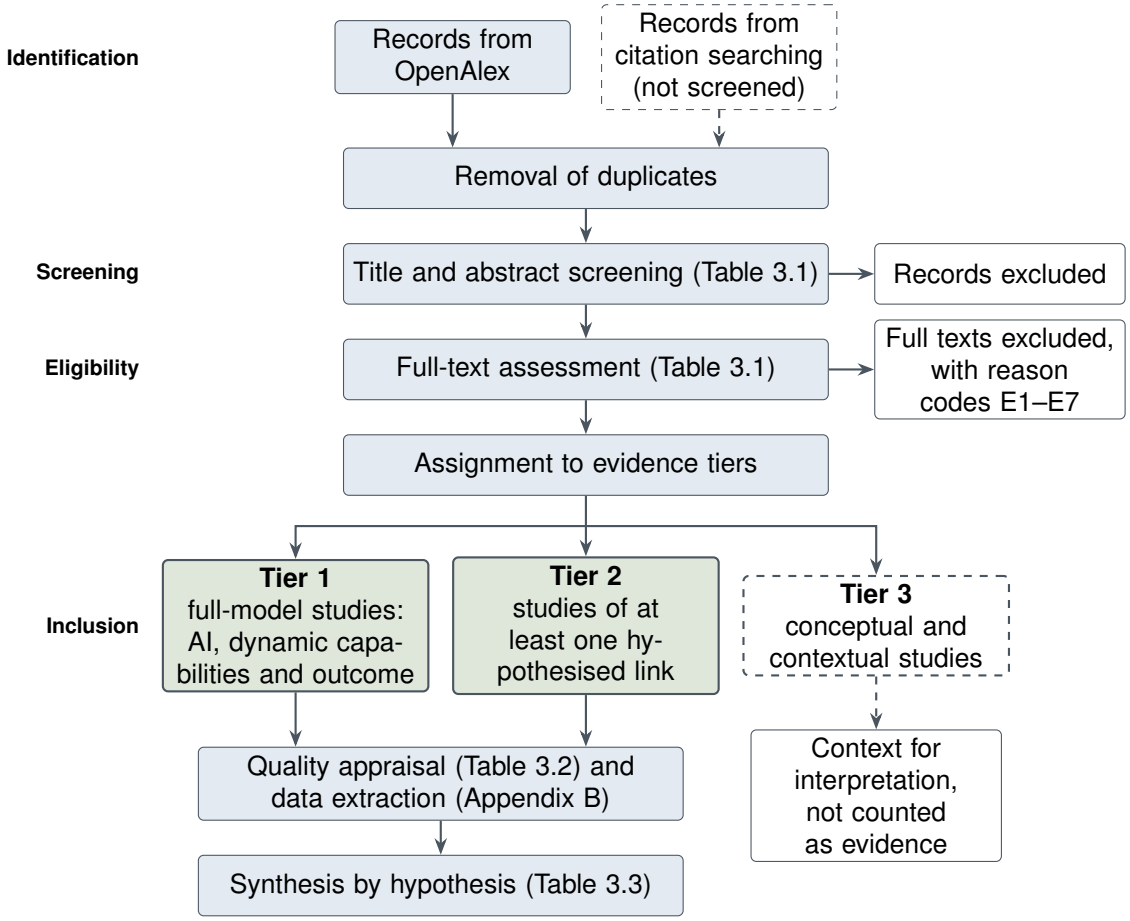
be appraised on the same basis as published findings. Such material may be used as context, but it is not counted as evidence. Second, the construct criterion is broader than the hypotheses on the side of AI but narrower on the side of outcomes. Studies of AI integration, assimilation or use are included, but classified as indirectly relevant evidence (Section 3.3.2), so that their inclusion does not overstate the support for the hypotheses. Studies that relate AI only to firm performance are excluded, because they examine neither the dependent variable nor the proposed mechanism of the thesis (Section 2.2.3); this criterion was fixed before screening began, after a scoping search showed that performance-only studies would otherwise dominate the evidence base. Studies that examine dynamic capabilities and firm performance together are retained, with performance treated as indirectly relevant for competitive advantage. A construct is treated as a dynamic capability when the study measures it and its authors explicitly define it as a dynamic capability or as one of its dimensions, as in the case of supply chain resilience or dynamic managerial capability. Four included studies meet the construct criterion only in this sense; they are marked in Appendix C, and their results are classified as related capabilities rather than as dynamic capabilities in the synthesis.

Screening proceeds in two stages, as shown in Figure 3.2. After duplicates have been removed, titles and abstracts are screened against the criteria, and the remaining records are assessed in full text. Every full-text exclusion is recorded with its reason code, so that the selection can be reported in a PRISMA 2020 flow diagram in Chapter 4 (Page et al., 2021). Following the PRISMA format for reporting does not by itself make a review systematic; the systematic character of the review rests on the procedures described in this chapter.

The included studies are assigned to three evidence tiers (Figure 3.2). Tier 1 contains full-model studies that examine an AI construct, dynamic capabilities and an outcome together. Tier 2 contains studies that examine at least one hypothesised link, such as the relationship between AI capability and dynamic capabilities. Tier 3 contains conceptual and contextual studies, which help to interpret the findings but are not counted as evidence for or against a hypothesis. Publications based on the same sample are linked and treated as one source of evidence.

Because the review is conducted by a single researcher, screening and data extraction were supported by a large language model (Claude, Anthropic), applied with written instructions that reproduce the criteria in Tables 3.1 and 3.2. At the title and abstract stage, each record was classified in two separate runs of the language model. Both runs used identical instructions, and neither had access to the other's output. No second human reviewer was involved, so the two runs are not independent reviewers in

Figure 3.2: Screening stages and evidence tiers



Source: author's own work; the stages follow the PRISMA 2020 flow diagram (Page et al., 2021).

the usual sense; they are two repeated classifications by the same model. The inclusion decision was then derived from each run's classification by fixed rules rather than by the model. Where the two runs led to different decisions, a third run of the model, with more detailed instructions, classified the record again, and its classification was used. At the full-text stage, every extraction was checked against the source in a separate verification run that corrected unsupported entries; every reported result had to be backed by a verbatim quotation from the accessible source text. All inclusion decisions and extracted results remain subject to checking by the author against the original papers; where only an abstract was accessible, such a check is limited to the abstract (Section 3.5).

3.3 Appraisal and Synthesis

Once the relevant studies have been identified, two further steps turn them into evidence. Quality appraisal establishes how far each study's results can be trusted, and synthesis combines the results across studies into a judgement on each hypothesis. The distinction matters because relevance and quality are different properties: a study

can address exactly the relationship of interest and still provide weak evidence, for example if its design cannot rule out alternative explanations. The following subsections explain how quality is assessed and data are extracted, and how the evidence is synthesised into judgements.

3.3.1 Quality Appraisal and Data Extraction

The weight of a review's conclusions depends on the quality of the studies it includes. In a review of 391 systematic reviews in management research, Rojon et al. (2021) found that authors paid more attention to explaining their protocol and search strategy than to the analysis and synthesis of the included studies' findings. This chapter therefore specifies the appraisal and the synthesis as precisely as the search.

Some guides for management reviews use lists of highly ranked journals as a basis for quality appraisal (Kraus et al., 2020). A journal's ranking says little about the design of an individual study, however, so this thesis appraises each study directly. The appraisal draws on the Mixed Methods Appraisal Tool (MMAT), which provides core quality criteria for qualitative, quantitative and mixed methods designs and can therefore be applied to both the survey and the case study evidence identified in Chapter 2 (Hong et al., 2018). These general criteria are supplemented by criteria that matter specifically for the hypotheses of this thesis. Table 3.2 lists the nine resulting criteria.

The criteria marked as key criteria in Table 3.2 address the main threats identified in Chapter 2. Measurement validity (Q3) matters because the studies define AI capability in different ways (Section 2.2.1); for H6, it also covers whether dynamic capabilities are correctly specified and validated as a higher-order construct (Sarstedt et al., 2019). Common method bias (Q4) matters because most studies collect predictors and outcomes from the same managers at the same time. Such bias has multiple sources, arises under widespread conditions and is not easy to fix (Podsakoff et al., 2024); statistical tests applied after data collection are therefore rated as partly met at most. The estimation of indirect effects (Q6) matters because mediation should be established by testing the indirect effect itself rather than by a sequence of separate significance tests (Zhao et al., 2010). Criteria Q5 and Q7 address temporal ordering and alternative explanations, because endogeneity threatens causal claims in management research (Hill et al., 2021) and standard mediation models rest on assumptions that cross-sectional data rarely satisfy (Rohrer et al., 2022). For qualitative studies, rigour of data collection and analysis (Q8) is the key criterion.

Each criterion is rated as met, partly met, not met or not applicable, and the ratings for every study are reported in Chapter 4. Rather than summing the ratings into a score, the overall quality of a study is derived from the explicit rule stated below Table 3.2.

Table 3.2: Quality appraisal criteria

No.	Criterion	Guiding question	Applies to	Basis
Q1	Research question and theory	Is the research question clearly stated and grounded in theory?	All studies	Hong et al. (2018)
Q2	Sampling	Is the sample appropriate for the research question, and are response rates and non-response bias reported?	Quantitative studies	Hong et al. (2018)
Q3*	Measurement validity	Are the constructs measured reliably and validly, and is it clear which AI construct is measured? Are higher-order constructs specified and validated correctly?	Quantitative studies	Hong et al. (2018); Sarstedt et al. (2019)
Q4*	Common method bias	Do predictors and outcomes come from different sources or points in time, or are other remedies applied and reported?	Survey studies	Podsakoff et al. (2024)
Q5	Temporal ordering	Are the measures separated in time, for example in a time-lagged or longitudinal design?	Quantitative studies	Hill et al. (2021); Rohrer et al. (2022)
Q6*	Indirect effects	Is the indirect effect estimated directly, for example with bootstrapped confidence intervals, rather than inferred from separate paths?	Studies of mediation	Zhao et al. (2010)
Q7	Alternative explanations	Are relevant control variables included, or methods used to address endogeneity?	Quantitative studies	Hill et al. (2021)
Q8*	Qualitative rigour	Are the data sources, the analysis and the interpretation adequate and coherent?	Qualitative and case studies	Hong et al. (2018)
Q9	Reporting	Are coefficients, measures of uncertainty and sample details reported in enough detail for extraction?	All studies	Author's addition

Rating: each criterion is rated as met, partly met, not met or not applicable. Overall quality is *high* if all applicable key criteria (*) are met and no more than one other applicable criterion is not met, *low* if any applicable key criterion is not met, and *moderate* otherwise.

Source: author's own work, based on the sources listed in the table.

Data are extracted with a standard form, which is shown in Appendix B. For each study, the form records the source and any linked publications; the country, the industry and whether the study concerns e-commerce or platform-based business; the design, sample and method of analysis; and the exact constructs and measures used. For each hypothesised path, it records the direction of the relationship, the coefficient or effect size, the reported uncertainty and whether the result is a direct association or an estimated indirect effect. The outcome is classified as firm performance, competitive advantage or the persistence of competitive advantage, as required by Section 2.2.3. The extracted data form the study matrix and the evidence matrix in Chapter 4.

3.3.2 Synthesis and Decision Rules

Because the constructs, contexts and designs of the included studies differ, their results are not pooled statistically. A narrative synthesis nevertheless requires explicit

methods. The Synthesis Without Meta-analysis (SWiM) guideline sets out nine reporting items for reviews in which effect estimates cannot be combined, including how studies are grouped, which synthesis method is used and how heterogeneity and the certainty of the evidence are addressed (Campbell et al., 2020). Although SWiM was developed for reviews of interventions, its items provide a useful structure for this thesis. Studies are grouped by hypothesis, the synthesis is based on the direction of the reported relationships and on whether an indirect effect was estimated, and differences in measurement, context and design are examined as possible explanations for divergent results.

Table 3.3 defines the categories and decision rules before they are applied. At study level, each result is classified as directly reported, indirectly relevant or contradictory (Panel A). At hypothesis level, the classified results are combined into one of five judgements, ranging from *supported* to *not yet examined* (Panel B). Five general rules govern this step (Panel C).

Three of the general rules address problems identified in Chapter 2. The independence rule ensures that studies based on the same sample, such as the two studies by Almheiri and colleagues, count only once (Section 2.3.2). The assignment rule separates H3 to H5 from H6, so that the same estimate never counts in support of both specifications (Section 2.4.2). The context rule keeps the conclusions within the scope of the thesis: because the hypotheses concern e-commerce and platform-based business, a hypothesis can only be judged *supported* if direct evidence of at least moderate quality comes from this context. Otherwise, the judgement is limited to *partly supported*, together with the note that support comes from other settings.

Where studies disagree, the differences are first examined in terms of measurement, context and design. A hypothesis is judged *inconclusive* only if the disagreement cannot be explained in this way and the conflicting studies are of similar quality. Each judgement is reported together with the quality ratings of the underlying studies, so that the reader can see both the direction and the strength of the evidence.

3.4 Research Quality and Ethics

The SRH guidelines require research to be transparent and repeatable. Mårtensson et al. (2016) describe the quality of research practice in four main areas: credible, contributory, communicable and conforming. These areas structure the quality measures of this thesis. Credibility is supported by the explicit criteria and decision rules, by the appraisal of every included study whose full text was accessible and by the second screening run. The contribution lies in assessing the evidence relationship by relationship, which addresses the research gap identified in Section 2.4.1. Communicability

Table 3.3: Evidence categories and decision rules

Panel A: Study-level evidence categories	
Directly reported	The study reports the hypothesised relationship using the constructs of the hypothesis (AI capability or AI competencies; dynamic capabilities or one of their dimensions; competitive advantage); for H3 to H6, it reports an estimated indirect effect.
Indirectly relevant	The study reports a related relationship, for example with AI integration, assimilation or use instead of AI capability, with firm performance instead of competitive advantage, with a mediator related to but not identical with a dimension of dynamic capabilities, or with mediation inferred from separate paths.
Contradictory	The study reports a null result or a relationship in the opposite direction.
No eligible evidence	No included study addresses the relationship.
Panel B: Hypothesis-level judgements	
Supported	At least two independent studies report the relationship directly and in the hypothesised direction; at least one of them is of moderate or high quality and comes from e-commerce or platform-based business; no contradictory direct evidence of equal or higher quality exists.
Partly supported	Supporting evidence exists but does not meet the criteria for *supported*, for example direct evidence from only one independent study, mainly indirectly relevant evidence or support only from other contexts.
Inconclusive	Supporting and contradictory direct evidence of similar quality exist, and the differences cannot be explained by measurement, context or design.
Contradicted	Contradictory direct evidence predominates among studies of moderate or high quality.
Not yet examined	No eligible evidence exists.
Panel C: General rules	
Independence	Publications based on the same sample count as one study.
Quality	Studies of low quality are reported, but cannot on their own establish *supported* or *contradicted*.
Outcome	Results for firm performance count as indirectly relevant for sustainable competitive advantage.
Assignment	Separate indirect effects for sensing, seizing or transforming count for H3 to H5; an overall indirect effect counts for H6; no estimate counts for both.
Context	Without direct evidence of at least moderate quality from e-commerce or platform-based business, the judgement is at most *partly supported*.

Source: author's own work, informed by Campbell et al. (2020) and Zhao et al. (2010).

is supported by reporting the selection in a PRISMA 2020 flow diagram (Page et al., 2021), by following the SWiM items for the synthesis (Campbell et al., 2020) and by documenting the search and the extraction in Appendices A and B. This documentation allows another researcher to repeat the review, in line with the aim of transparent and reproducible synthesis that underlies systematic reviews in management (Tranfield et al., 2003). Conformity is addressed by following the SRH guidelines and the ethical principles described below.

The thesis is based exclusively on published studies. No participants were contacted and no personal data were collected or processed, so the research does not raise the ethical issues associated with human participants. Its main ethical obligations concern

the treatment of the literature. Findings are reported accurately and completely, including null and contradictory results, and studies are not selected or interpreted in order to confirm the hypotheses. All sources are cited in accordance with the SRH guidelines, and the digital tools used are disclosed: the search was run through the OpenAlex API with the query documented in Appendix A, and screening and extraction were supported by a large language model as described in Section 3.2.2. Responsibility for every decision and every statement in the thesis remains with the author.

3.5 Methodological Limitations

The chosen design has several limitations. First, the review is conducted by a single researcher with the support of a language model. Two separate screening runs, fixed decision rules and a separate verification run reduce the risk of selection and extraction errors, but they cannot remove it. Because both runs come from the same model, they may share the same systematic misreadings, which a high agreement between them would not reveal. Second, the search is limited to one database, to journal articles in English and to the period from 2015 onwards, and the citation searching could not be completed. Relevant studies in other languages, in journals with incomplete records in OpenAlex or in the grey literature may therefore be missed. Because studies with significant results are more likely to be published, the restriction to journal articles may also favour supporting evidence, and this possibility is considered when the findings are interpreted. Third, the full text of many included studies is not openly accessible. For these studies, eligibility and data extraction rely on the abstract, and their methodological quality cannot be assessed. For nine included studies, the database record contained no abstract and no full text was accessible; they were retained provisionally on the basis of their titles, contribute no extracted results and are marked in Appendix C. Their eligibility remains to be confirmed against the full texts.

Fourth, the review cannot overcome the limitations of the primary studies. If most of them rely on cross-sectional surveys of single respondents, the review inherits their vulnerability to common method bias and endogeneity (Podsakoff et al., 2024; Hill et al., 2021). Its conclusions are therefore stated as associations, not as causal effects. Fifth, assigning heterogeneous constructs, such as agility or business model innovation, to the dimensions of dynamic capabilities requires judgement. These decisions are recorded in the extraction forms, so that they can be checked. Sixth, the synthesis is based mainly on the direction of effects. It gives less weight to their size and produces no new estimates of the relationships. Finally, the field is developing quickly, particularly with the spread of generative AI, so the evidence base may change soon after the search.

3.6 Chapter Summary

This chapter has described the research design of the thesis. Following a post-positivist and deductive logic, the thesis uses a structured literature review to assess how far published evidence supports the six hypotheses developed in Chapter 2. The review searches OpenAlex, applies explicit inclusion and exclusion criteria in two separate model-supported screening runs, appraises each study against nine quality criteria and synthesises the evidence narratively using predefined categories and decision rules.

These procedures are designed to make the review transparent and repeatable and to keep its conclusions within the limits of the underlying evidence. The following chapter applies them. It reports the results of the search and selection, describes the characteristics and quality of the included studies and assesses the evidence for each hypothesis.

4 Analysis and Discussion

This chapter applies the procedure described in Chapter 3. Section 4.1 reports the results of the search and selection and describes the included studies and their quality. Sections 4.2 and 4.3 assess the evidence for the direct and the mediating relationships, Section 4.4 discusses the findings in relation to the literature reviewed in Chapter 2, and Section 4.5 summarises the chapter.

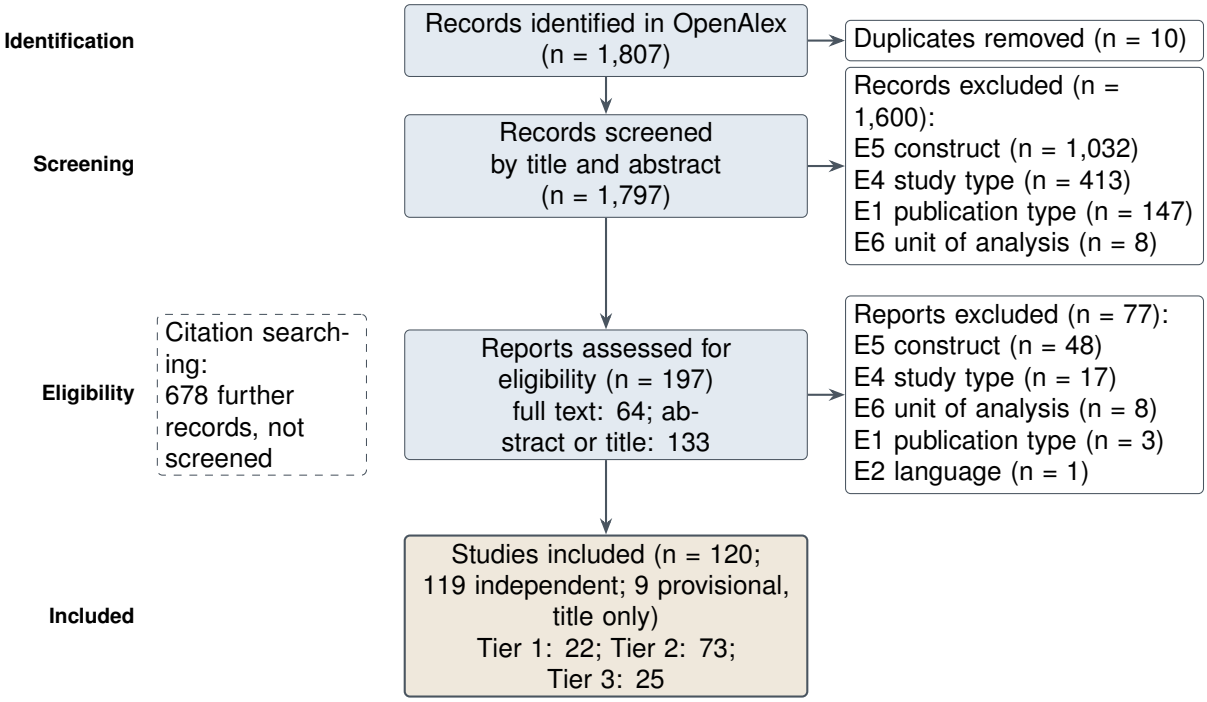
4.1 Overview of Included Studies

The search in OpenAlex on 22 October 2026 identified 1,807 records (Appendix A). Figure 4.1 shows how these records were reduced to the final evidence base. After the removal of 10 duplicates, 1,797 records were screened by title and abstract. The two screening runs of the language model agreed on 96.9% of the records, which corresponds to a Cohen's kappa of 0.84 for the include or exclude decision (Section 3.2.2). This figure shows that the model classified the records consistently across runs; it is not a measure of agreement between human reviewers. The 55 disagreements were resolved by a third run. At this stage, 1,600 records were excluded, most of them because they did not examine an organisational AI construct together with dynamic capabilities or competitive advantage (E5, 1,032 records), because they were literature reviews or similar publication types (E4, 413) or because they were not journal articles (E1, 147).

The remaining 197 reports were assessed for eligibility. An open-access full text could be retrieved for 64 of them; the other 133 were assessed on the basis of their abstracts or, where the database record contained no abstract, their titles. A further 77 reports were excluded at this stage, so that 120 studies were included. Of these, 28 were assessed in full text, 83 on their abstract and 9 on their title alone. The nine title-only studies are retained provisionally: they contribute no extracted results, and their eligibility still has to be confirmed against the full texts (Section 3.5). Two of the included studies, by Almheiri and colleagues, are based on the same sample of 344 public managers (Almheiri et al., 2024; Almheiri et al., 2025), so the 120 publications represent 119 independent studies, and the 95 empirical studies among them are based on 94 independent samples. The citation searching identified 678 further records, which could not be screened (Section 3.5). Appendix C lists all included studies.

The evidence base is very recent. As Figure 4.2 shows, 112 of the 120 studies were published in 2024 or later, and 57 appeared in 2026 alone, although that year was

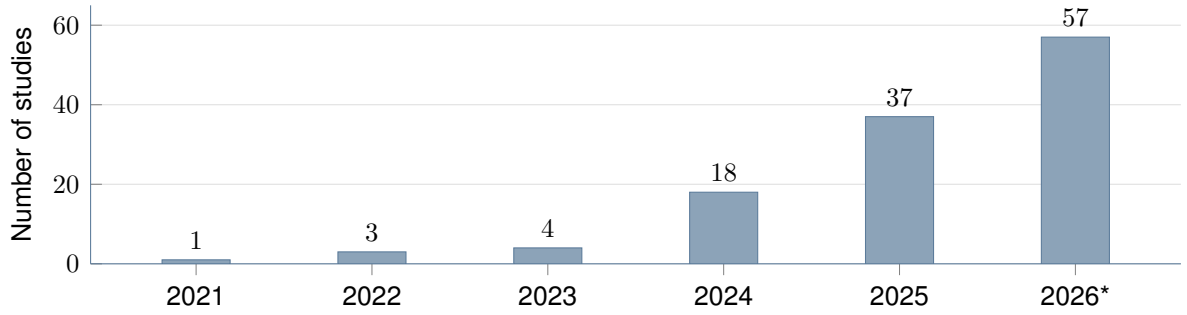
Figure 4.1: Flow of records through the review



Source: author's own work, following the PRISMA 2020 flow diagram (Page et al., 2021).

covered only until the date of the search. Research on AI capability, dynamic capabilities and competitive advantage is therefore still at an early and rapidly expanding stage, which matches the observation in Chapter 2 that much of the relevant work was published only recently.

Figure 4.2: Included studies by year of publication

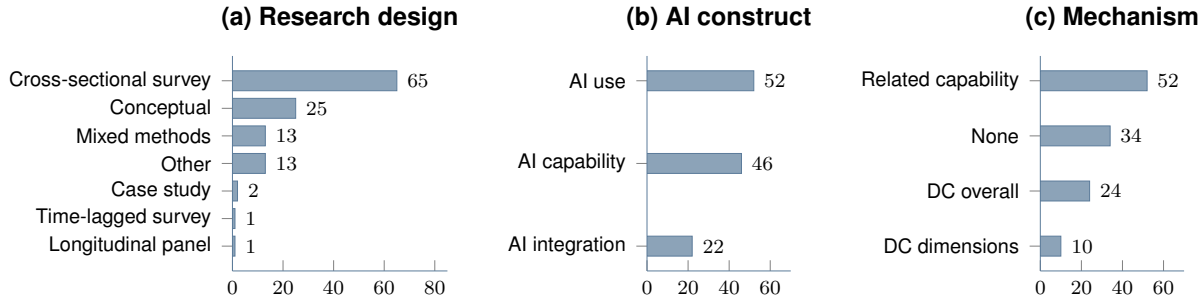


Source: author's own work, based on the review data; *2026 until 22 October. Years are the year of first publication recorded in OpenAlex, which can differ by one year from the issue year used in citations.

Figure 4.3 summarises the design of the included studies and the way they treat the two central constructs. Most empirical studies are cross-sectional surveys (65 of 95 empirical studies); only one study used a time-lagged design and one a longitudinal panel, and 25 studies are conceptual. The AI construct is most often the extent of AI use or adoption (52 studies), followed by AI capability (46) and AI integration (22). Dynamic capabilities are measured as an overall construct in 24 studies and through

their dimensions in only 10. In 52 studies, the mechanism is a related capability, such as agility or innovation capability, and 34 studies examine no capability mechanism at all. The outcome is competitive advantage in 74 studies and both competitive advantage and performance in another 20. Only one study measured whether an advantage persists over time (Dzreke, 2025). Thirty of the 95 empirical studies do not report the country of their sample in the accessible text; where it is reported, China (10 studies) is the most frequent context.

Figure 4.3: Design, AI construct and capability mechanism of the included studies



Source: author's own work, based on the review data (n = 120).

The publication pattern of the evidence is unusual. The 120 studies appeared in 105 different journals, and no journal published more than three of them. Established outlets of information systems and management research, such as *Industrial Marketing Management*, the *International Journal of Information Management* and the *Journal of Enterprise Information Management*, account for only a handful of studies, while most appeared in journals that published a single study on the topic. Such dispersion indicates that the field has not yet formed a cumulative research programme in which studies build on each other's measures and findings. It also means that the quality of the peer review behind individual studies varies considerably, which strengthens the case for appraising each study directly rather than relying on the reputation of the journal (Section 3.3.1).

The samples are of moderate size. Of the 80 empirical studies that report a sample size, the median is 312 respondents, and 16 studies are based on fewer than 150. Partial least squares structural equation modelling is by far the most common method of analysis (39 studies), followed by covariance-based structural equation modelling (9). Only 9 studies report that they collected data from more than one informant per organisation. In combination with the cross-sectional designs, this means that most results rest on the perceptions of a single manager, measured at a single point in time.

The methodological quality of the evidence is limited. Of the 95 empirical studies, 3 were rated high, 8 moderate and 9 low. For the other 75, the quality could not be assessed: 66 were accessible only as an abstract, 8 only as a title, and one full text

reported too little detail to be appraised. Among the 20 studies that could be appraised, the most frequent weakness was temporal ordering: only two studies separated their measures in time. The criterion on common method bias was met in 10 studies, and an indirect effect was estimated directly in 10. Other criteria were rarely met even where the full text was available. Only three studies fully met the sampling criterion, which requires an appropriate sampling frame and information on response rates and non-response bias, and only two fully met the criterion of controlling for alternative explanations. The problem is most acute where the evidence matters most. Of the 22 Tier 1 studies, which examine AI, dynamic capabilities and an outcome together, only four could be appraised: three were rated moderate and one low, while the quality of the remaining 18 could not be assessed. A rating of "not assessable" does not imply that a study is of low quality; it means that its quality is unknown. Because these studies nevertheless contribute most of the directly reported evidence, Sections 4.2.1 and 4.2.2 test whether the judgements change when only the appraisable studies are considered. Table 4.1 presents the key studies that inform the assessment of the hypotheses: all studies that report a hypothesised relationship directly, all studies of high or moderate quality, all empirical studies from e-commerce, platform or closely related digital contexts, all studies with contradictory results and one further study discussed in the text. The ratings of every included study are reported in Appendix C.

Table 4.1: Study matrix of key included studies

Study	Context; n	Design	AI construct	Mechanism	Outcome	Quality	Evidence
Studies reporting a hypothesised relationship directly							
Firmansyah et al. (2025)†	Digital MSMEs, Indonesia; 70	Survey, PLS-SEM	AI capability	–	Sustainable CA	Low	H1 D*
Shao et al. (2026)	SMEs, China; 546	Survey, SEM	AI capability	Business model innovation	CA	n.a.	H1 D; H6 I
Almereb et al. (2026)	Manufacturing, Jordan; 300	Survey, SEM	Generative AI	–	Cost and differentiation CA	n.a.	H1 D*
Kim & Seo (2026)	Exporters, South Korea; 400	Survey, PLS-SEM	AI capabilities	–	Differentiation and low-cost CA	n.a.	H1 D; H6 I
Giang et al. (2026)	SMEs, Vietnam; 463	Survey, PLS-SEM	AI capability	Organisational agility	CA	n.a.	H1 D*; H2 I; H6 I
Gafsi (2026)	Family firms, Tunisia; 120	Survey, SEM	AI capabilities	–	Competitive performance	n.a.	H1 D*
Al-Mamary (2025)	Entrepreneurial ventures; 327	Survey, PLS-SEM	AI capabilities	–	CA; venture success	n.a.	H1 D*
Tu et al. (2026)	Manufacturing, China; 304	Survey, bootstrap	AI capability	Learning; resilience	Competitiveness	n.a.	H1 D; H6 I
Waqas et al. (2024)	Manufacturing (CPEC); 463	Survey, PLS-SEM	AI capability	Green innovation	Green CA	n.a.	H1 D*; H2 I; H6 I
Almheiri et al. (2024); Almheiri et al. (2025)	Public sector, UAE; 344 (same sample)	Survey, scale development	AI capability	DC overall; creativity	Organisational performance	n.a.	H1 I; H2 D; H6 I
De Fano et al. (2026)	SMEs, Europe; 102	Survey, PLS-SEM	AI ambidexterity	DC dimensions	SME performance	n.a.	H2 D; H6 I
Neira-Fernández et al. (2026)	Firms, Spain; n.r.	Survey, PLS-SEM	AI readiness	Dynamic digital capabilities	Digital maturity	n.a.	H2 D; H6 I

Table 4.1 continued

Study	Context; n	Design	AI construct	Mechanism	Outcome	Quality	Evidence
Studies of high or moderate quality							
Texeira (2026)	Restaurants, Brazil; 400	Survey, bootstrap	AI capability	AI-enabled sustainability capabilities	Sustainable CA	High	H1 I; H2 I; H6 I
Wang et al. (2026)	High-tech firms, China; 229	Time-lagged survey	AI capabilities	Disruptive innovation	Sustainable competitive performance	High	H1 C; H2 I
Guo et al. (2025)	Cross-border platform sellers, China; 322	Survey, CB-SEM	AI adoption with strategic agility	International marketing capabilities	Innovation and brand advantage	High	H1 I; H2 I; H6 I
Malik (2021)†	Technology-intensive firms; 385	Survey, PLS-SEM	AI capability	Innovation and organisational capability	CA	Moderate	H2 I; H6 I
Srahin et al. (2026)	Multiple sectors, Jordan; 287	Survey, PLS-SEM	Augmented analytics adoption	Firm agility	CA	Moderate	H1 I; H2 I; H6 I
Dzreke (2025)†	Listed firms, North America and Europe; 1,200	Panel and interviews	BDA–AI integration	Orchestration capability	Competitive persistence	Moderate	H1 I; H6 I
Li & Sattayatham-rongthian (2026)	Tourism, China; 502	Survey, CB-SEM	AI orientation	Absorptive capacity; opportunity recognition	Performance	Moderate	H2 I; H4 I; H5 I
Noor et al. (2026)†	E-commerce, Pakistan; n.r.	Survey, PLS-SEM	Generative AI adoption	Market sensing; content capabilities	Marketing performance	Moderate	H1 I; H2 I; H3 I; H6 I
Deng et al. (2025)	Audit firms, multinational; 312	Survey, CB-SEM	Generative AI adoption	Adaptation mechanisms	Audit performance	Moderate	H1 I; H2 I; H6 I
Adamu et al. (2026)	Digital sectors, Nigeria; 268	Survey, fsQCA	AI capability	–	CA	Moderate	H1 I
Nguyen & Nguyen (2026)	Start-ups, Vietnam; 315	Survey, PLS-SEM	AI adoption intensity (moderator)	DC overall	Start-up performance	Moderate	–
Further studies from e-commerce contexts, studies with contradictory results and other studies discussed in the text							
Awamleh & Bustami (2022)	E-commerce, Jordan; 224	Survey, PLS-SEM	AI use	IT capabilities	CA	n.a.	H6 I
Bunea & Corboș (2026)	E-retailing; n.r.	Mixed methods	AI implementation	AI benefits; customer experience	Cost and differentiation CA	n.a.	H2 I; H6 I
Mashi et al. (2024)	Manufacturing SMEs, Nigeria; 346	Survey, PLS-SEM	AI adoption and use facets	–	CA	Low	H1 mixed
Chen et al. (2026)	Logistics, USA; 318	Survey, CB-SEM	AI infrastructure	Green innovation	Sustainable CA	n.a.	H1 C; H2 I; H6 I
Sofiyan et al. (2026)	Naval units, Indonesia; 81 units	Multilevel survey	AI capability	Organisational change capacity	Performance	n.a.	H2 C
Mukhopadhyay et al. (2024)	FMCG, India; 26 interviews	Qualitative	AI-enabled marketing	DC; marketing capabilities	CA	n.a.	H1 I; H2 I
Kokash (2026)	Tourism; n.r.	Survey, CB-SEM and fsQCA	AI capabilities	DC overall	Financial and market performance	n.a.	H1 I; H6 C

Abbreviations: CA = competitive advantage; DC = dynamic capabilities; D = directly reported; D* = directly reported, classification provisional (see Section 4.2.1); I = indirectly relevant; C = contradictory; n.a. = not assessable (abstract only); n.r. = not reported; † = publication details or peer-review status not yet confirmed.

Source: author's own work, based on the review data.

4.2 Direct Relationships

The conceptual model rests on two direct relationships. H1 links AI capability to sustainable competitive advantage, and H2 links AI capability to dynamic capabilities. The

second relationship is a precondition for the mediating mechanism: if AI capability were not related to dynamic capabilities, the proposed indirect paths through sensing, seizing and transforming could not exist. Both relationships are assessed as associations reported in the included studies, since the cross-sectional designs that dominate the evidence base do not allow causal conclusions. The following subsections assess the evidence for each relationship in turn.

4.2.1 AI Capability and Sustainable Competitive Advantage (H1)

Nine independent studies report a positive relationship between AI capability and competitive advantage directly. They cover SMEs in China (Shao et al., 2026) and Vietnam (Giang et al., 2026), manufacturing firms in Jordan, China and the China–Pakistan Economic Corridor (Almereb et al., 2026; Tu et al., 2026; Waqas et al., 2024), Korean exporters (Kim & Seo, 2026), Tunisian family businesses (Gafsi, 2026), entrepreneurial ventures (Al-Mamary, 2025) and digital micro, small and medium-sized enterprises in Indonesia (Firmansyah et al., 2025). The last study is the only one of the nine from a digital business context; it reports a path coefficient of 0.361 between AI capability and sustainable competitive advantage, but in a sample of only 70 firms, with low methodological quality, in a journal whose peer-review status could not be confirmed, and with a sample description that its accessible abstract does not repeat. The quality of the other eight studies could not be assessed, because only their abstracts were accessible. For six of the nine, moreover, the classification as directly reported is provisional. Two report AI capability only jointly with other digital capabilities, so that its specific contribution is not visible in the accessible results (Giang et al., 2026; Waqas et al., 2024); one measures generative AI, which is classified as a capability only on the basis of the abstract's wording (Almereb et al., 2026); in one, competitive advantage may be a dimension of AI's effect rather than its outcome (Al-Mamary, 2025); and one names competitive performance as its dependent variable but reports its finding as an effect on firm performance (Gafsi, 2026); the sixth is the small Indonesian study. Only three studies report an AI-specific relationship with competitive advantage in the accessible text without such a qualification (Shao et al., 2026; Kim & Seo, 2026; Tu et al., 2026).

A further 44 studies provide indirectly relevant evidence, and among them are the strongest studies of the review. In a high-quality survey of 322 Chinese firms that sell through cross-border platforms, the combination of AI adoption and strategic agility was positively related to product innovation advantage ($\beta = 0.519$) and brand upgrading advantage ($\beta = 0.336$) (Guo et al., 2025). Both relationships also ran through international marketing capabilities, so that the coupling was related to the advantages directly as well as indirectly. In a survey of 287 respondents in Jordan, the adoption

of augmented analytics had a total effect of 0.542 on competitive advantage, but the direct effect became small once firm agility was taken into account; its bootstrapped confidence interval included zero, although the reported p-value (0.036) would indicate significance (Srahin et al., 2026). A high-quality study of 400 decision-makers in Brazilian restaurants reports a similar pattern: AI capability was not significantly related to sustainable competitive advantage once AI-enabled sustainability capabilities were included as mediators (Texeira, 2026). In these two studies, AI relates to competitive advantage mainly through other capabilities, whereas in the platform-seller study both the direct and the indirect paths were significant.

Three studies report results that contradict H1. In a time-lagged survey of 229 Chinese high-tech firms, AI capabilities showed no significant main effect on sustainable competitive performance ($b = 0.05$); their role was rather to strengthen the effect of social entrepreneurship orientation on disruptive innovation (Wang et al., 2026). In that study, AI capabilities were modelled mainly as a moderator and were measured in the same survey wave as disruptive innovation, so that the null coefficient comes from a moderation model rather than from a test designed for H1. A study of 318 logistics firms in the United States found that AI infrastructure did not achieve sustainable performance or competitive advantage directly, but only through green innovation, and two of the routes from AI infrastructure through green innovation were not significant either (Chen et al., 2026). A survey of Nigerian manufacturing SMEs found mixed results: AI adoption was not significantly related to competitive advantage, whereas the extent of AI applications was (Mashi et al., 2024).

The studies also differ in what they mean by competitive advantage. Some distinguish cost-based from differentiation-based advantage (Almereb et al., 2026; Kim & Seo, 2026), some measure specific advantages, such as product innovation and brand upgrading (Guo et al., 2025), and others focus on green competitive advantage (Waqas et al., 2024; Chen et al., 2026) or on competitive performance relative to rivals (Gafsi, 2026; Wang et al., 2026). Almost all of these measures ask managers to rate their firm's position in comparison with competitors at the time of the survey. They therefore capture a perceived advantage at one point in time, whereas the resource-based view defines a sustained advantage by its persistence despite competitors' attempts to imitate it (Barney, 1991; Peteraf, 1993). Even the directly reported evidence on H1 thus concerns competitive advantage rather than sustainable competitive advantage in the sense of this thesis.

Two further analyses show how fragile the direct evidence is. The first concerns the AI construct. All nine directly reported results come from studies that measure AI capability, whereas the 28 studies that measure the extent of AI use provide only indirectly

relevant evidence on H1. The largest part of the literature therefore answers a different question from the one asked in this thesis: whether firms that use AI more intensively report an advantage, not whether a firm-specific AI capability produces one. The second analysis is a sensitivity check that considers only the 20 studies whose quality could be appraised. In this subset, only one study reports the relationship directly, and it is of low quality (Firmansyah et al., 2025); among the eleven studies of high or moderate quality, seven provide indirectly relevant evidence and one contradicts H1 (Wang et al., 2026). The direct support for H1 thus rests almost entirely on studies that could not be appraised.

The overall direction of the evidence also deserves a critical comment. Of the 70 results on the relationship between AI and competitive advantage that were extracted from the included studies, 57 are positive, 7 null, 5 mixed and only 1 negative. Such a one-sided distribution is consistent with a real positive association, but it is also what would be expected if studies with non-significant results were less likely to be published. The restriction of the evidence base to journal articles (Section 3.2.2) cannot rule out this possibility, so the high share of positive results should not be read as strong confirmation.

According to the decision rules in Table 3.3, H1 is *partly supported*. Direct supporting evidence exists, but none of it comes from a study of moderate or high quality, the classification of six of the nine studies is provisional, and the one high-quality study coded as contradictory, in which AI capabilities were modelled mainly as a moderator, found no significant main effect. The judgement would be the same if only the three unqualified studies were counted as direct evidence. The context rule leads to the same judgement: the only directly reported evidence from a digital business context is a small study of low quality (Firmansyah et al., 2025).

4.2.2 AI Capability and Dynamic Capabilities (H2)

Three independent samples report a positive relationship between AI capability and dynamic capabilities directly. AI capability was positively related to dynamic capabilities in a sample of public organisations in the United Arab Emirates (Almheiri et al., 2024; Almheiri et al., 2025), AI ambidexterity fostered dynamic capabilities, measured as sensing, seizing and transforming, in 102 European SMEs (De Fano et al., 2026), and AI readiness was related to dynamic digital capabilities in Spanish firms (Neira-Fernández et al., 2026). None of these samples comes from e-commerce, and the quality of all three could not be assessed from the accessible text.

The transferability of this evidence to the focal context is limited for different reasons in each case. The sample of Almheiri and colleagues consists of government organisations, which do not compete in markets in the way that e-commerce firms do, so that their dynamic capabilities serve public service performance rather than competitive advantage (Almheiri et al., 2024; Almheiri et al., 2025). The study of De Fano et al. (2026) examines SMEs that participate in a European digital innovation hub, a group that is likely to be more digitally advanced than the average firm, and Neira-Fernández et al. (2026) relate AI readiness to digital maturity rather than to an advantage over competitors. The direct evidence for H2 thus comes from settings in which AI may build dynamic capabilities, but in which the link to competition, which is central to e-commerce, plays little role.

The 42 studies with indirectly relevant evidence relate AI to capabilities that resemble one dimension of dynamic capabilities. AI capability was strongly related to innovation capability ($\beta = 0.68$) and organisational capability ($\beta = 0.62$) in 385 technology-intensive organisations (Malik, 2021), a study whose publication details could not be confirmed, augmented analytics adoption to firm agility ($\beta = 0.625$) (Srahin et al., 2026), and AI orientation to absorptive capacity ($\beta = 0.312$) among 502 managers of Chinese tourism businesses (Li & Sattayathamrongthian, 2026). The only e-commerce study with an explicit dynamic capability dimension found that generative AI adoption was strongly related to market sensing ($\beta = 0.571$) in Pakistani e-commerce firms (Noor et al., 2026). Qualitative studies also link AI and dynamic capabilities, although without establishing a direction: interviews in Indian consumer goods firms describe how dynamic capabilities, exercised through AI technologies, enhance marketing practices (Mukhopadhyay et al., 2024), AI-powered customer relationship management capabilities could be mapped onto the micro-foundations of dynamic capabilities (Alnofeli et al., 2025), and a Delphi panel of 14 experts mapped AI and other digital technologies onto sensing, seizing and transforming in manufacturing (Ghobakhloo et al., 2025).

Two results qualify this picture. In 81 naval units in Indonesia, AI capability had no direct effect on organisational change capacity; the effect appeared only under transformational leadership (Sofiyan et al., 2026). More fundamentally, the direction of the relationship is not settled. One study modelled dynamic capabilities as an antecedent of generative AI adoption rather than as its outcome ($\beta = 0.453$) (Deng et al., 2025), and a study of Vietnamese start-ups treated AI adoption as a moderator of the effect of dynamic capabilities on performance (Nguyen & Nguyen, 2026). Because almost all studies are cross-sectional, their data cannot distinguish between these specifications (Rohrer et al., 2022). The sensitivity check is even more revealing for H2 than for H1. If only the 20 appraisable studies are considered, no study reports the relationship between AI capability and dynamic capabilities directly; all 12 relevant studies in this

subset, 8 of them of high or moderate quality, relate AI to capabilities that resemble dynamic capabilities rather than to dynamic capabilities themselves. The direct evidence for H2 therefore comes exclusively from studies that could only be judged on their abstracts. This does not make that evidence wrong, but it means that the most important link of the conceptual model has not yet been demonstrated in a study whose methods could be examined.

H2 is therefore judged *partly supported*: the direct evidence is consistent but comes from three samples of unknown quality outside the focal context.

4.3 Mediating Relationships

Before the individual mediators are assessed, it is useful to look at the ten included studies that explicitly address sensing, seizing and transforming as dimensions (Figure 4.3). Their designs show where the literature stands. Four of them are conceptual: they propose that AI supports sensing, seizing and transforming (Karami, 2026), discuss how AI substitutes for or complements human intelligence in each dimension (Siaw & Ali, 2025), or develop similar typologies (two further papers listed in Appendix C). One derives such links from the judgement of a Delphi panel of experts (Ghobakhloo et al., 2025). Two are qualitative or mixed-methods studies that identify AI-related capabilities for each dimension from expert interviews (Feroz & Kwak, 2024) or from interview and survey material (Maune, 2026). Only three are quantitative (De Fano et al., 2026; Neira-Fernández et al., 2026; Noor et al., 2026), and none of them reports an indirect effect through a single dimension on competitive advantage. A further quantitative study names sensing, seizing and reconfiguring but models dynamic capabilities only as an overall antecedent of AI adoption (Deng et al., 2025). Theorising about the dimensions of dynamic capabilities thus runs well ahead of testing.

4.3.1 Sensing Capability (H3)

No study estimates the indirect effect of AI capability on competitive advantage through sensing. The closest evidence comes from the e-commerce study discussed above, which reports a significant serial indirect effect of generative AI adoption through market sensing and brand visibility on entrepreneurial marketing performance (0.110) (Noor et al., 2026). Because the AI construct is adoption rather than capability, the outcome is marketing performance rather than competitive advantage and the path runs through an additional mediator, this result is indirectly relevant only. De Fano et al. (2026) measured sensing as part of dynamic capabilities, but the accessible results do not separate the dimensions. Because the study itself models market sensing as the sensing dimension and estimates the indirect effect through it separately, the result

counts for H3 under the assignment rule (Table 3.3). H3 is therefore judged *partly supported*, on the basis of a single indirectly relevant study.

The non-quantitative studies add a qualification that matters for the interpretation of H3. A mixed-methods study of 312 respondents in Zimbabwe reports that AI enhanced sensing through automated monitoring, real-time data integration and predictive analytics (Maune, 2026). Conceptual work, however, suggests that the role of AI may differ between internal data analytics and the scanning of the external environment through which opportunities are sensed (Siaw & Ali, 2025); the abstract of that paper is not consistent about which of the two AI suits better, so the precise proposition still has to be checked against its full text. If such a distinction holds, the contribution of AI to sensing depends on what is sensed. In e-commerce, where much relevant market information is generated internally through customer transactions and behaviour, AI may support sensing more strongly than in industries where opportunities arise outside the firm's own data. This proposition has not been tested, but it suggests that H3 may hold more strongly in the focal context of this thesis than elsewhere.

4.3.2 Seizing Capability (H4)

No included study reports an indirect effect through seizing measured as a dimension of dynamic capabilities. One study frames absorptive capacity as a seizing capability and reports a significant indirect effect of AI orientation on firm performance through it (0.067) (Li & Sattayathamrongthian, 2026). Because the AI construct is an orientation rather than a capability, the outcome is firm performance rather than competitive advantage, and absorptive capacity is related to, but not identical with, seizing as defined in Table 2.1, this result is indirectly relevant only. Under the assignment rule, a separate indirect effect through a mediator that the study itself assigns to one dimension counts for the hypothesis on that dimension and not for H6; this is the same treatment as the market sensing result for H3. H4 is therefore judged *partly supported*, on the basis of a single indirectly relevant study.

4.3.3 Transforming Capability (H5)

The same applies to transforming. The study of Chinese tourism businesses frames entrepreneurial opportunity recognition as a transforming pathway, with an indirect effect of 0.057 on firm performance (Li & Sattayathamrongthian, 2026), and the Delphi study maps AI onto transforming activities without testing an effect (Ghobakhloo et al., 2025). Neither result measures transforming as a dimension of dynamic capabilities, and no study estimates an indirect effect through it on competitive advantage. The first result is nevertheless indirectly relevant and, under the assignment rule, counts for

H5. H5 is therefore judged *partly supported*, on the basis of a single indirectly relevant study. The serial indirect effect through both mediators in the same study (0.009) spans two dimensions and is not assigned to any single hypothesis.

4.3.4 Dynamic Capabilities as a Higher-Order Construct (H6)

No study reports an estimated indirect effect of AI capability on competitive advantage through dynamic capabilities as defined in this thesis, so there is no directly reported evidence for H6. The 51 studies with indirectly relevant evidence fall into two groups. The first group estimates indirect effects on competitive advantage through capabilities related to, but not identical with, dynamic capabilities: innovation capability (0.31) and organisational capability (0.27) (Malik, 2021), firm agility (0.445, with only a small remaining direct effect) (Srahin et al., 2026), international marketing capabilities (0.346 and 0.242) (Guo et al., 2025), AI-enabled sustainability capabilities, whose indirect effect increased with the firms' sustainable food innovation orientation (Texeira, 2026), organisational learning and resilience (Tu et al., 2026) and orchestration capability (Dzreke, 2025). The second group examines dynamic capabilities themselves, but with firm performance as the outcome or without an estimated indirect effect (Almheiri et al., 2024; De Fano et al., 2026; Neira-Fernández et al., 2026).

Across all included studies, the related capabilities that serve as mediators can be grouped by their content. Innovation-related capabilities are the most frequent (24 studies), followed by learning and knowledge capabilities, including absorptive capacity (14), agility and flexibility (12), marketing and customer capabilities (8), resilience (7), orchestration and integration capabilities (5) and business model innovation (3). Mapping these groups onto the three dimensions of dynamic capabilities is possible only in part. Agility combines the sensing of change with a quick response and therefore spans sensing and seizing; absorptive capacity covers the acquisition as well as the transformation of knowledge; and innovation capability can be read as seizing or as transforming, depending on whether the emphasis lies on exploiting an opportunity or on renewing the resource base. Because most of these constructs cut across the dimensions, the evidence collected under H6 cannot simply be redistributed to H3 to H5; only where a study itself assigns a mediator to a single dimension is the result counted for that dimension (Sections 4.3.1 to 4.3.3).

The way in which mediation is established is a further reason for caution. Of the studies that report a mediating mechanism, only 13 estimate the indirect effect directly, for example with bootstrapped confidence intervals, whereas 38 claim mediation on the basis of separate significant paths or without a clearly reported test. As Zhao et al. (2010) argue, mediation is demonstrated by the indirect effect itself, so the claims of

these 38 studies cannot be accepted at face value. Even where indirect effects were estimated, they rest on cross-sectional data, and standard mediation models require assumptions about the order and independence of the variables that such data rarely satisfy (Rohrer et al., 2022). The consistent pattern of indirect effects is therefore best read as a robust association between AI, related capabilities and competitive advantage, not as proof of a causal chain.

Where indirect effects were estimated, their size varies considerably. In the study of Jordanian firms, the indirect effect of augmented analytics adoption on competitive advantage through firm agility was 0.445 and accounted for most of the total effect of 0.542 (Srahin et al., 2026). In the study of Chinese tourism businesses, by contrast, the indirect effects of AI orientation on firm performance through absorptive capacity and opportunity recognition were 0.067 and 0.057, and the serial effect through both was only 0.009 (Li & Sattayathamrongthian, 2026). In the restaurant study, the indirect effect through AI-enabled operational sustainability rose from 0.086 at a low to 0.155 at a high level of sustainable food innovation orientation (Texeira, 2026). Differences of this magnitude reflect different constructs, outcomes and estimation methods, which is one reason why the results cannot be pooled. They also suggest that the strength of the capability pathway depends on the context and on which capability is examined, a point that the synthesis by direction of effect (Section 3.3.2) cannot capture.

Three observations limit what these studies show. First, some of them merge the constructs that the thesis keeps apart. Ayoub & Sopuru (2026) measure an "AI-enabled dynamic capability", and Shao et al. (2026) interpret AI capability itself as a dynamic capability, so that their results cannot show whether dynamic capabilities mediate the effect of AI. Second, one study explicitly tested the mediating role of dynamic capabilities between AI capabilities and performance in tourism firms and did not find it (Kokash, 2026). Third, none of the 120 included studies specifies dynamic capabilities as a higher-order construct. Under the decision rules, indirectly relevant evidence suffices for the judgement *partly supported*, and H6 is therefore judged *partly supported*. The basis of this judgement must be stated precisely, however: the evidence consistently shows that AI capability is associated with competitive advantage through other organisational capabilities, but not specifically through dynamic capabilities as a higher-order construct, and the only explicit test of mediation by dynamic capabilities found none.

4.4 Discussion

The preceding sections reported what the evidence shows for each hypothesis. This section asks what these results mean. It addresses three questions: how the findings

relate to the literature reviewed in Chapter 2, what the evidence says specifically about e-commerce and platform-based business, and how the conceptual model needs to change in the light of the evidence. Answering these questions requires interpretation beyond the decision rules, and the judgements made here should therefore be read as the author's interpretation rather than as results of the review procedure.

4.4.1 Comparison with Prior Literature

Table 4.2 brings the judgements together. None of the six hypotheses can be judged *supported*. All six are *partly supported* under the decision rules, but the basis of this judgement differs greatly between them. H1 and H2 rest on several directly reported results, almost all from studies of unknown quality; H3, H4 and H5 each rest on a single indirectly relevant result, two of them from the same study; and H6 rests on studies of related capabilities rather than of dynamic capabilities as a higher-order construct. The label *partly supported* therefore spans very different amounts of evidence, which is why Table 4.2 states the basis of each judgement. The evidence is thus broad but shallow: more than 100 studies relate AI to capabilities and competitive advantage, but few test the specific mechanism proposed in Chapter 2, few can be appraised, and almost none come from e-commerce or platform-based business.

The most consistent finding concerns the way AI relates to competitive advantage. In two of the appraised studies, the direct association between AI and competitive advantage became small or non-significant once other capabilities were included, while the indirect association through those capabilities remained (Srahin et al., 2026; Texeira, 2026), and a study of AI infrastructure that could be judged only on its abstract reports a similar pattern (Chen et al., 2026). The pattern is not uniform: in the high-quality study of platform sellers, the direct and the indirect paths were both significant (Guo et al., 2025), and other studies report a direct association alongside indirect ones (Tu et al., 2026; Kokash, 2026). The pattern resembles what Mikalef et al. (2020) found for big data analytics, whose effect on competitive performance was fully mediated by dynamic capabilities. It also fits the resource-based argument developed in Section 2.1.1: AI is valuable, but it is rarely rare or difficult to imitate on its own, and an advantage arises only when AI is combined with firm-specific capabilities (Barney, 1991; Kemp, 2024). The results thus support the central intuition of the thesis, that AI capability works at least partly through organisational capabilities, even though they do not yet support the specific role of dynamic capabilities.

The mediators found in the literature are, however, remarkably diverse. Agility, innovation capability, absorptive capacity, organisational learning, marketing capabilities and orchestration capability all appear as links between AI and competitive advantage. This

Table 4.2: Evidence matrix and judgements for H1 to H6

H	Directly re-ported	Indirectly relevant	Contra-dictory	Quality of direct evidence	Focal context	Judgement
H1	9*	44	3	1 low, 8 not assessable	1 direct (low), 3 indirect	Partly supported
H2	3	42	1	3 not assessable	3 indirect	Partly supported
H3	0	1	0	–	1 indirect	Partly supported (one indirect study)
H4	0	1	0	–	–	Partly supported (one indirect study)
H5	0	1	0	–	–	Partly supported (one indirect study)
H6	0	51	1	–	4 indirect	Partly supported (related capabilities only)

Numbers are independent studies; one study can provide evidence for several hypotheses, and a study with results in two categories is counted once, in the stronger category. *For six of the nine studies, the classification as directly reported is provisional (Section 4.2.1). The indirectly relevant evidence for H4 and H5 comes from the same study. Focal context: e-commerce, platform-based and closely related digital business.

Source: author's own work, applying the decision rules in Table 3.3.

diversity is consistent with the view of Eisenhardt & Martin (2000) that dynamic capabilities take the form of specific, identifiable processes, but it also confirms the measurement problem identified in Section 2.2.2. Studies rarely state how their mediators relate to sensing, seizing and transforming, and only 10 of the 120 studies address these dimensions at all, only three of them quantitatively. The dimension-specific mechanism proposed by Teece (2007) has therefore hardly been tested. The one study that distinguished the dimensions in the earlier literature, Selgas-Cors & Thiébaut (2026), was not retrieved by the database search; even it reports only an overall indirect effect.

The evidence also adds a qualification that the conceptual model did not anticipate: the effect of AI depends on boundary conditions. In restaurants, the indirect effect of AI capability increased with the firms' sustainable food innovation orientation (Texeira, 2026), the effect on organisational change capacity depended on transformational leadership (Sofiyan et al., 2026), and environmental and technological turbulence shaped the role of AI in two further studies (Al Dhaheri et al., 2024; Kokash, 2026). In other studies, AI was itself the moderator, strengthening the effect of dynamic capabilities (Nguyen

& Nguyen, 2026) or of entrepreneurial orientation (Wang et al., 2026). These findings suggest that AI may amplify existing capabilities as much as it creates new ones, a possibility that is consistent with the complementation dynamics described by Krakowski et al. (2023).

A further check concerns the studies that Chapter 2 identified as most relevant (Table 2.3). Of these 15 studies, 4 are included in the review (Shao et al., 2026; Almheiri et al., 2024; Almheiri et al., 2025; Al Dhaheri et al., 2024). Another four were retrieved but excluded because they relate AI only to performance (Chen et al., 2022; Mikalef et al., 2023a; Mikalef et al., 2023b; Neiroukh et al., 2025), which reflects the protocol decision described in Section 3.2.2. The remaining seven were not retrieved by the search. For six of them, the reason lies in their outcome or construct, such as servitization, creativity, innovation capability, business risk or perceived user value, which falls outside the outcome terms of the search string (Abou-Foul et al., 2023; Ameen et al., 2024; Gao et al., 2025; Drydakis, 2022; Yu et al., 2026) or, in one case, outside AI altogether (Mikalef et al., 2020). The seventh, Selgas-Cors & Thiébaut (2026), examines AI integration, the three dimensions of dynamic capabilities and competitive advantage and would have qualified; its absence shows that a single database and a title-and-abstract search can miss relevant studies, particularly in smaller journals. The completed citation search would have been the appropriate remedy.

The findings can also be compared with the earlier review of the AI business value literature by Enholm et al. (2022), which distinguished first-order effects of AI, at the level of processes, from second-order effects at the level of the organisation. The present review suggests how the two levels are connected. The studies rarely find that AI produces competitive advantage, a second-order effect, on its own; instead, AI first strengthens capabilities such as agility, innovation or learning, and these capabilities are associated with the advantage. In this sense, the review adds a mechanism to that earlier distinction, although the cross-sectional evidence cannot show that the sequence actually unfolds over time. The comparison with Wamba-Taguimdje et al. (2020) points in the same direction: their conclusion that AI creates value only when processes are reconfigured is consistent with the finding that the indirect paths carry most of the effect.

The review also speaks to the theoretical debate about how AI can produce a competitive advantage at all. Kemp (2024) argues that AI creates an advantage only when it is situated in the firm through specific organisational activities, and Moser et al. (2024) respond that situatedness must be taken even more seriously in theorising about AI-based advantage. The empirical studies support the general thrust of this argument,

because they locate the effect of AI in firm-specific capabilities rather than in the technology. They do not, however, measure situatedness itself. Most studies treat AI capability as a set of resources and skills that can be rated on a questionnaire, which says little about how AI is embedded in the routines of a particular firm. Closing this gap would require designs that observe how AI is used in organisations over time, for example longitudinal case studies.

Finally, the quality of the evidence limits all of these conclusions. Most studies are cross-sectional surveys of single respondents, so their results are vulnerable to common method bias (Podsakoff et al., 2024) and cannot establish the direction of the relationships, as the competing specifications of Deng et al. (2025) and Nguyen & Nguyen (2026) illustrate. Only one study measured the persistence of an advantage (Dzreke, 2025), and its reporting raises doubts about its data. As Brynjolfsson et al. (2021) show, the economic effects of AI may only appear after complementary investments have been made, which cross-sectional performance measures cannot capture. The judgements in Table 4.2 should therefore be read as statements about the state of the evidence rather than as statements about the true effect of AI.

4.4.2 Evidence from E-Commerce and Platform Contexts

Because the research questions concern e-commerce and platform-based business, this section separates the evidence from this focal context from the evidence that has to be transferred from other industries. Only 7 of the 120 included studies come from this focal and closely related context, and only four of them are empirical studies of e-commerce or platform-based business, one of them of complementors; their designs and quality differ considerably. Direct evidence about platform owners is almost entirely absent. One high-quality survey examines 322 Chinese firms that sell through cross-border platforms, that is, complementors (Guo et al., 2025). Three studies concern e-commerce firms or retailers: a moderate-quality survey of Pakistani e-commerce firms (Noor et al., 2026), a survey of 224 respondents from Jordanian e-commerce enterprises (Awamleh & Bustami, 2022) and a mixed-methods study of electronic retailing (Bunea & Corbos, 2026). Two studies examine digitally active firms more broadly: a moderate-quality configurational study of 268 firms in Nigeria (Adamu et al., 2026) and a low-quality survey of 70 micro, small and medium-sized enterprises in Indonesia, described in the extraction as digital firms, a description that the accessible abstract does not repeat (Firmansyah et al., 2025). The only study of platform owners is conceptual (Dzreke & Dzreke, 2025).

Considered on their own, these studies support the following statements. For H1, the small Indonesian study of low quality reports a direct positive relationship between AI

capability and sustainable competitive advantage, and three studies provide indirectly relevant evidence: AI adoption combined with strategic agility relates to innovation and brand advantages among platform sellers (Guo et al., 2025), generative AI adoption relates to marketing performance in e-commerce firms (Noor et al., 2026), and AI capability and cloud infrastructure can substitute for each other in configurations associated with competitive advantage (Adamu et al., 2026). For H2, the platform-seller and e-commerce studies just mentioned and the study of electronic retailing (Bunea & Corboș, 2026) relate AI to capabilities that resemble dynamic capabilities, namely international marketing capabilities, market sensing and content capabilities, and the benefits of AI implementation; none measures dynamic capabilities as such. For H3, the Pakistani e-commerce study is the only evidence on sensing in the entire review. For H4 and H5, there is no evidence from the focal context; the only indirectly relevant study comes from tourism (Li & Sattayathamrongthian, 2026). For H6, four studies report that related capabilities, such as information technology capabilities, international marketing capabilities or customer experience, mediate the relationship between AI and competitive advantage: the platform-seller, Pakistani and electronic retailing studies, together with the Jordanian e-commerce survey (Awamleh & Bustami, 2022).

Within the focal context alone, the judgements would therefore be largely the same as for the whole review, but on a much narrower basis: H1, H2, H3 and H6 would be partly supported, each by between one and four studies, whereas H4 and H5 would have to be judged not yet examined, because no evidence on them comes from this context. Wherever Sections 4.2 and 4.3 draw on studies from manufacturing, public organisations, tourism or SMEs in general, their conclusions involve a transfer of evidence to the focal context. Such a transfer is plausible for the general logic that AI works through complementary capabilities, because this pattern recurs across very different settings. It is much less plausible for the mechanisms that distinguish platforms from other firms, as the following paragraphs explain.

The review thus confirms the context gap identified in Section 2.4.1. None of the focal studies tests dynamic capabilities as a mediator of competitive advantage, and the mechanisms that are specific to platforms, such as data network effects (Gregory et al., 2021) and the integrative capabilities of platform leaders (Helfat & Raubitschek, 2018), have not been examined empirically at all; the only included study on platform owners is conceptual and of uncertain peer-review status (Dzreke & Dzreke, 2025).

The geographical distribution of the evidence reinforces this gap. Of the 95 empirical studies, 30 do not report their country in the accessible text. Of the others, most come from Southeast Asia (14), East Asia (13) and the Middle East (12), followed by

Africa and South Asia; only three studies each come from North America and Europe. The evidence base is therefore dominated by firms in emerging markets, many of them SMEs. This is valuable, because these firms are often under-represented in management research. It also means, however, that the evidence says little about the large platform firms of the United States, Europe and China, for which data network effects and ecosystem orchestration are most relevant (Gregory et al., 2021; Rietveld & Schilling, 2021). For a thesis concerned with e-commerce and platform-based business, this is a significant limitation of the available evidence.

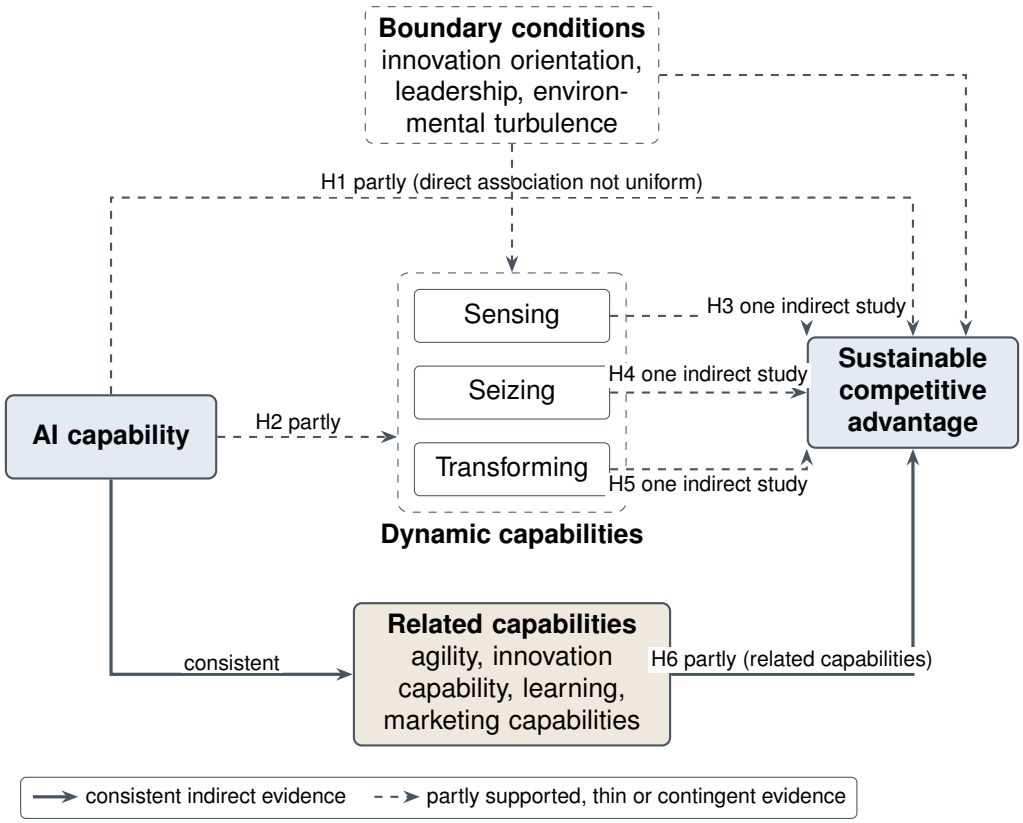
There are nevertheless some indications that the focal context matters. In a panel of 1,200 listed firms, the effect of combining big data analytics and AI on competitive persistence was reported to be stronger in fast-moving sectors such as FinTech and e-commerce than in capital-intensive sectors such as heavy manufacturing (Dzreke, 2025). If this result holds, AI-based advantages may be both more attainable and more contested in e-commerce than elsewhere: the speed of these markets rewards the capabilities that AI supports, but it also allows rivals to catch up quickly. Because of the doubts about the reporting of that study noted in Section 4.4.1, the result can only be treated as a hypothesis for future research, but it is consistent with the view of Cennamo (2021) that competition in digital markets follows its own logic.

4.4.3 Revised Conceptual Model

Figure 4.4 revises the conceptual model of Figure 2.3 in the light of the evidence. Three changes are made. First, related organisational capabilities, such as agility, innovation capability and organisational learning, are added as the pathway that the literature actually tests; it carries most of the indirectly relevant evidence for H6. Second, the dimension-specific paths through sensing, seizing and transforming are marked as resting on a single indirectly relevant study each, none of them from a test of the dimension on competitive advantage. Third, boundary conditions are added, because several studies show that the effect of AI depends on conditions such as innovation orientation, leadership and environmental turbulence.

The revised model has two implications for the following chapter. Theoretically, it suggests that the link between AI and competitive advantage is better described by a broad set of capabilities that AI strengthens than by the three dimensions of dynamic capabilities alone. Methodologically, it shows where evidence is missing: dimension-specific tests of sensing, seizing and transforming with competitive advantage as the outcome, longitudinal designs that can capture the persistence of an advantage, and studies of e-commerce retailers and platform owners.

Figure 4.4: Revised conceptual model



Source: author's own work, based on the evidence in Table 4.2.

4.5 Chapter Summary

The structured review identified 1,807 records and included 120 studies, almost all of them published since 2024. AI capability is consistently associated with competitive advantage, but mostly indirectly, through capabilities such as agility, innovation capability and organisational learning. No hypothesis is supported, and all six are partly supported on very different bases: the direct relationship with competitive advantage (H1) and the relationship with dynamic capabilities (H2) on several studies of mostly unknown quality, the mediating roles of sensing, seizing and transforming (H3 to H5) on a single indirectly relevant study each, and the mediating role of dynamic capabilities as a whole (H6) on studies of related capabilities rather than of dynamic capabilities themselves.

These judgements are limited by the quality of the evidence: most studies are cross-sectional, the quality of most could not be assessed, and only four empirical studies concern e-commerce or platform-based business. The revised model therefore treats related capabilities and boundary conditions as part of the mechanism and marks the paths that remain untested. The following chapter draws conclusions from these findings and answers the research questions.

5 Conclusion

This chapter concludes the thesis. Section 5.1 summarises the main findings, and Section 5.2 answers the four research questions. Section 5.3 discusses the theoretical and managerial implications, and Section 5.4 reflects critically on the limitations of the thesis and outlines directions for future research.

5.1 Summary of Findings

The thesis set out to evaluate how far published empirical evidence supports the proposition that AI capabilities contribute to a sustainable competitive advantage in e-commerce and platform-based businesses through dynamic capabilities. Chapter 1 argued that this question matters because AI tools are becoming widely accessible, so that any lasting advantage is more likely to depend on how firms organise around AI than on the technology itself. Building on the resource-based view and dynamic capabilities theory, Chapter 2 developed a conceptual model in which AI capability relates to sustainable competitive advantage directly (H1), relates to dynamic capabilities (H2), and works through sensing, seizing and transforming individually (H3 to H5) and through dynamic capabilities as a whole (H6). Chapter 3 described a structured literature review with explicit criteria, quality appraisal and decision rules to assess this model.

The review in Chapter 4 identified 1,807 records and included 120 studies. Its findings can be condensed into four key messages. First, the literature is new and growing fast: 112 of the included studies were published in 2024 or later. Second, AI is consistently associated with competitive advantage, but mainly indirectly. In two of the appraised studies, the direct association of AI became small or non-significant once other capabilities, such as agility or AI-enabled sustainability capabilities, were included in the model (Srahin et al., 2026; Texeira, 2026), although other studies, including the high-quality study of platform sellers, report direct and indirect associations side by side (Guo et al., 2025). Third, the specific mechanism proposed in this thesis has hardly been tested. Only 10 of the 120 studies address the dimensions of dynamic capabilities, only three of them quantitatively, and no study estimates separate indirect effects through sensing, seizing and transforming on competitive advantage. Fourth, the evidence from e-commerce and platform-based business is very thin: only four empirical studies concern this context, and the only one of high quality examines complementors that sell through cross-border platforms rather than platform owners or retailers (Guo et al., 2025).

Measured against the decision rules defined in Table 3.3, no hypothesis meets the criteria for *supported*, and all six are at most partly supported. The basis of this judgement differs greatly: H1 and H2 rest on several directly reported results from studies of mostly unknown quality, H3, H4 and H5 on a single indirectly relevant result each, and H6 on studies of related capabilities rather than of dynamic capabilities as a higher-order construct. This outcome does not mean that the proposed relationships are absent. It means that the published evidence is not yet strong, specific or context-relevant enough to confirm them.

The thesis makes three contributions. First, it provides a relationship-by-relationship assessment of a literature that has grown so quickly that its overall state is difficult to see: by separating the direct link, the link to dynamic capabilities and each mediating path, it shows that the support for the overall argument is much stronger than the support for its individual steps. Second, it makes the basis of each judgement explicit. The decision rules, the distinction between directly reported and indirectly relevant evidence, and the sensitivity checks allow readers to see exactly why a hypothesis is judged as it is and how the judgement would change if the evidence changed. Third, it develops a revised conceptual model that adds related capabilities and boundary conditions to the original model and marks the paths that remain untested, which turns the gaps in the evidence into a research agenda.

5.2 Answers to the Research Questions

RQ1: To what extent does published empirical evidence support a relationship between AI capabilities and sustainable competitive advantage?

The evidence supports this relationship only in part. Nine independent studies report a positive association between AI capability and competitive advantage directly, although for six of them this classification is provisional, and 44 further studies report related evidence, for example with AI use instead of AI capability or with competitive advantage reached through other capabilities (Section 4.2.1). The direct evidence, however, comes from studies whose quality is low or could not be assessed, and the one high-quality study coded as contradictory, in which AI capabilities were modelled mainly as a moderator, found no significant main effect (Wang et al., 2026). In two appraised studies and one study judged on its abstract, AI relates to competitive advantage mainly through mediating capabilities (Srahin et al., 2026; Texeira, 2026; Chen et al., 2026), but this is not the case in every study (Guo et al., 2025). In addition, only one study measured whether an advantage persists over time (Dzreke, 2025). The answer is also sensitive to the quality of the evidence: among the studies whose methods could be appraised, only one reports the relationship directly, and the AI construct that most

studies measure is the extent of AI use rather than AI capability. The answer to RQ1 is therefore that published evidence supports an association between AI and competitive advantage, but not a direct relationship with an advantage that is demonstrably sustainable.

RQ2: To what extent does published empirical evidence support a relationship between AI capabilities and dynamic capabilities?

The evidence again supports this relationship in part. Three independent samples report directly that AI capability is positively related to dynamic capabilities (Almheiri et al., 2025; De Fano et al., 2026; Neira-Fernández et al., 2026), and 42 studies relate AI to capabilities that resemble one dimension of dynamic capabilities, such as agility, innovation capability, absorptive capacity or market sensing (Section 4.2.2). The direction of the relationship is, however, unresolved. Some studies treat dynamic capabilities as an antecedent of AI adoption (Deng et al., 2025), and others treat AI as a moderator of the effect of dynamic capabilities (Nguyen & Nguyen, 2026). Because almost all studies are cross-sectional, they cannot decide between these specifications. Moreover, all directly reported evidence comes from studies that could only be judged on their abstracts, and from settings such as the public sector, in which competition plays little role. The answer to RQ2 is that AI and dynamic capabilities are consistently associated, but the evidence does not show that AI capability builds dynamic capabilities rather than the reverse.

RQ3: To what extent does published empirical evidence indicate that sensing, seizing and transforming capabilities each mediate the relationship between AI capabilities and sustainable competitive advantage?

The published evidence hardly addresses this question. For sensing, a single study of e-commerce firms reports an indirect effect of generative AI adoption on marketing performance through market sensing (Noor et al., 2026), which makes H3 partly supported on the basis of one indirectly relevant study. For seizing and transforming, a single study of tourism businesses reports indirect effects of AI orientation on firm performance through absorptive capacity and opportunity recognition, which its authors interpret as seizing and transforming mechanisms (Li & Sattayathamrongthian, 2026); under the same rule, H4 and H5 are therefore partly supported on the basis of one indirectly relevant study each. No study estimates an indirect effect through sensing, seizing or transforming, measured as dimensions of dynamic capabilities, on competitive advantage. Of the ten studies that address the three dimensions explicitly, four are conceptual, one is a Delphi study and two are qualitative or mixed-methods, so that theorising about the dimensions clearly runs ahead of testing. The answer to RQ3

is therefore that the dimension-specific mediation proposed by dynamic capabilities theory (Teece, 2007) has not yet been tested in a way that allows a conclusion.

RQ4: To what extent does published empirical evidence indicate that dynamic capabilities, as a higher-order construct, mediate this relationship?

The evidence indicates that organisational capabilities mediate the relationship, but not specifically dynamic capabilities as a higher-order construct. Fifty-one studies provide indirectly relevant evidence, most of them estimating indirect effects through related capabilities such as innovation capability, agility, international marketing capabilities or organisational learning (Malik, 2021; Srahin et al., 2026; Guo et al., 2025; Tu et al., 2026). No study estimates the indirect effect of AI capability on competitive advantage through dynamic capabilities as defined in this thesis, none specifies dynamic capabilities as a higher-order construct, one study finds no mediation by dynamic capabilities (Kokash, 2026), and some studies merge AI capability and dynamic capabilities into a single construct (Ayoub & Sopuru, 2026; Shao et al., 2026). In addition, only 13 studies estimate an indirect effect directly, whereas 38 infer mediation from separate paths. H6 is therefore partly supported only in the sense defined by the decision rules, through related capabilities. The answer to RQ4 is that the mediating logic of the conceptual model is broadly consistent with the evidence, but that the mediator is better described as a family of capabilities strengthened by AI than as dynamic capabilities in the narrow sense.

5.3 Implications

The findings of a review have implications for theory and for practice, but these implications have to match the strength of the evidence. Because none of the hypotheses is fully supported, the implications of this thesis are stated conditionally: they describe what follows if the observed patterns hold, and they identify where theory and practice should be cautious. The following subsections discuss the theoretical implications first and the managerial implications second.

5.3.1 Theoretical Implications

The findings have three implications for theory. First, they support the resource-based argument that AI on its own is unlikely to be a source of sustained advantage. Because AI is valuable but widely accessible, it rarely meets the criteria of rareness and inimitability (Barney, 1991). The finding of several studies that its direct association weakens once firm-specific capabilities are considered is consistent with this view and with the argument that AI must be situated in the organisation to create an advantage (Kemp, 2024). It also mirrors earlier evidence on big data analytics, whose effect

on competitive performance was fully mediated by dynamic capabilities (Mikalef et al., 2020).

Second, the findings qualify the role of dynamic capabilities in this mechanism. The literature does not test sensing, seizing and transforming as distinct mediators; it tests a wide range of more specific capabilities. This is consistent with the view that dynamic capabilities take the form of identifiable processes, such as product development or strategic decision making (Eisenhardt & Martin, 2000), but it leaves open whether these processes add up to the higher-order capability described by Teece (2007). Future theorising should therefore specify how particular capabilities, such as agility or absorptive capacity, relate to the three dimensions, instead of using dynamic capabilities as a general label.

Third, the findings suggest that AI may amplify existing capabilities as much as it creates new ones. Several studies found that AI strengthened the effect of other capabilities or orientations rather than acting directly (Nguyen & Nguyen, 2026; Wang et al., 2026), and that its effect depended on innovation orientation, leadership or environmental turbulence (Texeira, 2026; Sofiyan et al., 2026; Kokash, 2026). This interpretation fits the idea that AI changes the sources of competitive advantage through complementation with existing human and organisational capabilities (Krakowski et al., 2023). The revised model in Figure 4.4 therefore adds related capabilities and boundary conditions to the original model.

A fourth implication concerns the clarity of the constructs. The review found that AI capability, AI use and AI integration are often treated as interchangeable, that some studies define AI capability as a dynamic capability in its own right (Shao et al., 2026; Ayoub & Sopuru, 2026), and that dynamic capabilities are frequently invoked as a theoretical lens without being measured. Under these conditions, a positive association between “AI” and “capabilities” can mean very different things. Theoretical progress therefore depends less on new hypotheses than on agreed definitions and measures: AI capability as a firm-specific bundle of resources (Mikalef & Gupta, 2021), dynamic capabilities as the higher-order capacity to sense, seize and transform (Teece, 2007), and competitive advantage as a position relative to competitors that persists over time (Barney, 1991). The debate about situated AI (Kemp, 2024; Moser et al., 2024) adds that the embedding of AI in organisational routines may be the decisive feature, and this feature is not captured by current survey measures at all.

5.3.2 Managerial Implications

For managers in e-commerce and platform-based businesses, the main implication is that investment in AI technology is unlikely to produce a lasting advantage on its own.

The evidence indicates that AI creates value through capabilities that surround it, such as the ability to act quickly on new information, to innovate and to learn (Srahin et al., 2026; Malik, 2021; Tu et al., 2026). Managers should therefore assess AI projects not only by their technical performance but also by whether they strengthen these organisational capabilities, and they should invest in the processes, skills and structures that allow AI-generated insights to be turned into decisions and changes.

A second implication concerns the conditions under which AI pays off. The indirect effect of AI increased with a sustainable innovation orientation in one study of restaurants (Texeira, 2026) and depended on transformational leadership in another (Sofiyan et al., 2026). Leadership attention and a strategic orientation towards innovation are therefore part of an AI strategy rather than separate issues. For firms that sell through platforms, the evidence on sellers using cross-border platforms suggests that AI is most effective when it is combined with strategic agility and marketing capabilities (Guo et al., 2025).

The implications differ between types of firms in digital markets. Platform owners control the data generated on their platforms and can therefore use AI to strengthen data network effects, but research suggests that this advantage depends on legitimacy, data stewardship and user-centric design (Gregory et al., 2021). AI capability can even reduce the value users perceive if it raises concerns about privacy (Yu et al., 2026), and opaque, poorly governed AI can lead to poor decisions and competitive disadvantage (Rana et al., 2022). For platform owners, AI governance is therefore part of the competitive strategy, not merely a compliance task. E-commerce retailers, by contrast, compete with rivals who can buy similar AI tools, so their advantage depends on how well AI is combined with their own customer knowledge, marketing capabilities and capacity to change (Noor et al., 2026; Bunea & Corboș, 2026). Complementors that sell through platforms rely partly on the platform's AI; for them, the evidence suggests that combining AI with their own strategic agility and marketing capabilities is what creates an advantage over other sellers (Guo et al., 2025).

Finally, managers should treat claims about the competitive benefits of AI with caution. Most of the evidence comes from cross-sectional surveys of managers' perceptions, often outside e-commerce, and the effects of AI may take time to appear in performance, because AI requires complementary intangible investments (Brynjolfsson et al., 2021). Firms that monitor the effects of their AI investments over several years, and compare them with competitors, will be better placed to judge whether an advantage is lasting.

5.4 Limitations and Future Research

The conclusions of this thesis must be read in the light of its limitations, which were set out in Section 3.5. The most important are that the review relied on a single database, that the citation searching could not be completed, that the quality of 75 of the 95 empirical studies could not be assessed, mostly because only their abstracts were accessible, that nine included studies could be judged only on their titles, and that screening and extraction were supported by a language model. The thesis also inherits the limitations of the primary studies, most of which are cross-sectional surveys of single respondents and therefore vulnerable to common method bias and reverse causality (Podsakoff et al., 2024; Hill et al., 2021). The judgements in Chapter 4 are therefore statements about the current state of the published evidence, not about the true effect of AI. Repeating the review with additional databases, the completed citation search and full access to the included studies would strengthen these judgements.

A critical reflection on the research process adds three further points. First, the decision to exclude studies that relate AI only to firm performance was made after a scoping search and before screening. It kept the review focused on the dependent variable and the mechanism of the thesis, but it also removed a large body of evidence on the performance effects of AI, including well-known studies (Mikalef & Gupta, 2021; Chen et al., 2022), so a broader review might have reached somewhat different conclusions on H1. Second, the use of a language model made it possible to screen almost 1,800 records twice and to require a verbatim quotation from the accessible source text for every extracted result, which would have been difficult for a single researcher. At the same time, it introduces a new kind of risk, because the model may misread studies in systematic ways that are harder to detect than individual human errors; the manual checks described in Section 3.2.2 are therefore essential. Third, the decision rules proved strict. Because the quality of most studies could not be assessed, no hypothesis could reach the judgement *supported*. This strictness was intended, because it prevents the volume of weak evidence from being mistaken for strength, but it also means that the judgements describe the limits of the evidence as much as the relationships themselves.

These limitations, together with the gaps identified in Chapter 4, point to five directions for future research. First, studies should measure sensing, seizing and transforming as distinct dimensions and estimate the indirect effect through each of them, so that H3 to H5 can be tested directly. Higher-order models of dynamic capabilities should be specified and validated with established procedures (Sarstedt et al., 2019). Second, research needs longitudinal or time-lagged designs, ideally combining survey data with archival measures of performance over several years, to establish the direction of the

relationships and the persistence of an advantage. Third, the constructs need to be kept apart: AI capability should be measured separately from dynamic capabilities and from AI adoption, so that their relationships can be examined rather than assumed.

Fourth, the context of e-commerce and platform-based business deserves dedicated study. Mechanisms that are specific to platforms, such as data network effects (Gregory et al., 2021) and the integrative capabilities through which platform leaders orchestrate their ecosystems (Helfat & Raubitschek, 2018), have not yet been examined empirically, and the differences between platform owners, retailers and complementors remain unexplored (Cennamo, 2021). Fifth, the boundary conditions found in this review, such as innovation orientation, leadership and environmental turbulence, should be built into future models from the start. Qualitative case studies of platform firms could help to identify how these conditions shape the conversion of AI capability into a lasting advantage.

Two further directions follow from the critical discussion in Chapter 4. Research should give more attention to the negative and null effects of AI. Only one of the 70 extracted results on AI and competitive advantage was negative, which is unlikely to reflect the full range of experiences with AI in practice, given the risks of opaque and poorly governed AI (Rana et al., 2022). Registered reports, studies that explicitly test for failure and the publication of null results would give a more balanced picture. Finally, the rapid spread of generative AI raises the question of whether the findings of studies conducted before its diffusion still apply. Generative AI lowers the cost of many capabilities that were previously firm-specific, such as content creation and customer communication, and may therefore make AI-based advantages even harder to sustain. Future research should examine whether generative AI capabilities follow the same indirect logic as earlier forms of AI or whether they erode the advantages that firms have built.

In conclusion, the thesis finds that AI capability is associated with competitive advantage mainly through the organisational capabilities that AI strengthens, but that the evidence is not yet sufficient to confirm that dynamic capabilities, and in particular sensing, seizing and transforming, carry this effect in e-commerce and platform-based businesses. Answering this question will require research that is more specific in its constructs, longer in its time horizon and closer to the context of digital platforms.

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Appendix A Search Log

This appendix documents the search described in Section 3.2.1. The query was run through the public interface of OpenAlex and searched titles and abstracts with word stemming. Table A.1 reports the exact query, the limits, the date and the number of records.

Table A.1: Search query and search log

Source	Search query (titles and abstracts)	Limits	Date	Records
OpenAlex	`("artificial intelligence" OR "machine learning" OR "AI") AND (capability OR capabilities OR sensing OR seizing OR reconfiguring OR reconfiguration) AND ("competitive advantage" OR "competitive advantages" OR "firm performance" OR "organizational performance" OR "organisational performance" OR "competitive performance")`	Type article; language English; publication date from 1 January 2015 (upper filter value 31 December 2026, see note); no subject filter	22 October 2026	1,807
Citation searching	Works cited by, or citing, the 121 studies eligible before the final language check, filtered with the AI block AND (capability block OR outcome block)	As above	22 October 2026	678 new, not screened

Note: The search was run on 22 October 2026, so it can only contain records available in OpenAlex on that date. The upper date filter was set to the end of 2026 so that articles already published online but dated to a later 2026 issue were not excluded; it does not extend the search beyond the search date.

Source: author's own work.

The benchmark test retrieved Mikalef & Gupta (2021), Chen et al. (2022) and Shao et al. (2026). Applying a filter for business, management, decision sciences, economics and social sciences reduced the result to 1,188 records but excluded Chen et al. (2022), which OpenAlex assigns to computer science; the filter was therefore not used.

Appendix B Data Extraction Form

This appendix shows the form used to extract data from each included study (Section 3.3.1). One form is completed per study. Results for different hypothesised paths are recorded in separate rows of the results section, so that each path can be classified on its own.

Table B.1: Data extraction form

Section	Field	Content
Identification	Study ID	Running number used in Chapter 4
	Reference	Full reference
	Linked publications	Other publications based on the same sample
Context	Country and industry	As reported by the study
	Focal context	E-commerce retailer, platform owner, complementor or other
Design	Design	Cross-sectional survey, time-lagged or longitudinal survey, case study or mixed methods
	Sample	Size, unit and respondents (single or multiple informants)
	Analysis	For example PLS-SEM, covariance-based SEM, regression or qualitative analysis
Constructs	AI construct	Name, definition, dimensions and source of the measure
	Dynamic capabilities	Overall or by dimension; first-order or higher-order specification
	Outcome	Firm performance, competitive advantage or persistence of advantage; measure used
Results (one row per path)	Path and hypothesis	For example AI capability $\rightarrow$ sensing (H2); indirect effect via sensing (H3)
	Direction and size	Sign and coefficient or effect size
	Uncertainty	Confidence interval, standard error or p-value
	Type of result	Direct association, estimated indirect effect or mediation inferred from separate paths
Appraisal	Criteria Q1–Q9	Rating per criterion (Table 3.2)
	Overall quality	High, moderate or low
Synthesis	Evidence tier	Tier 1, 2 or 3 (Figure 3.2)
	Evidence category	Per hypothesis (Table 3.3, Panel A)
	Notes	Mapping decisions, for example the assignment of a mediator to a dimension of dynamic capabilities

Source: author's own work.

Appendix C Included Studies

This appendix lists all 120 studies included in the structured review (Section 3.2.2), with their evidence tier (Figure 3.2) and overall quality rating (Table 3.2). Studies that are discussed in Chapter 4 are also listed in the bibliography. Studies that are listed in the bibliography are cited with the year of their issue; for the other studies, and in Figure 4.2, the year is the year of first publication recorded in OpenAlex. Markers: T = retained provisionally on the basis of the title alone (no abstract or full text accessible; eligibility to be confirmed); E = meets the construct criterion only because the authors define a measured construct as a dynamic capability (Section 3.2.2); † = publication details or peer-review status not yet confirmed.

Table C.1: Studies included in the review

Study	Title, journal and identifier	Tier	Quality
Kabir et al. (2025)	Deep Learning-Enabled Business Models for Competitive Advantage. *International Journal on Science and Technology*. https://doi.org/10.71097/ijsat.v16.i1.1824	3	–
Abdelfattah et al. (2025)	Harnessing Artificial Intelligence, Business Intelligence, and Digital Technologies for achieving supply chain excellence in Oman: Investigating the mediating role of predictive analytics. *Journal of Open Innovation Technology Market and Complexity*. https://doi.org/10.1016/j.joitmc.2025.100644	2	n.a.
Adamu (2026)	Digital Resource Theory: A Configurational Extension Of The Resource-Based View For Ai, Data, And Cloud Environments. *African Journal of Social and Behavioural Sciences*. OpenAlex W7220572703	3	–
Adamu et al. (2026)	From Resource Possession To Resource Synergy: A Configurational Test Of Digital Resource Theory Using Fsqca. *GUSAU Journal of Business Administration*. https://doi.org/10.57233/gujoba.v5i1.17	2	moderate
Afrianty et al. (2026)	Navigating competitive advantage through digital leadership and AI adoption: evidence from large business organisations in Indonesia. *Cogent Business & Management*. https://doi.org/10.1080/23311975.2026.2681244	2	n.a.
Alhelal et al. (2025)	AI-Driven Sustainable Competitive Advantage in Tourism and Hospitality: Mediating Roles of Digital Culture and Skills. *Sustainability*. https://doi.org/10.3390/su17198903	2	n.a.
Almereb et al. (2026)	The influence of green supply chain management on competitive advantage: the role of generative artificial intelligence. *Management & Sustainability An Arab Review*. https://doi.org/10.1108/msar-07-2025-0239	2	n.a.
Almheiri et al. (2024)	Artificial intelligence capabilities, dynamic capabilities and organizational creativity: contributing factors to the United Arab Emirates Government's organizational performance. *Journal of Modelling in Management*. https://doi.org/10.1108/jm2-11-2022-0272	1	n.a.
Almheiri et al. (2025)	Examining the impact of artificial intelligence capability on dynamic capabilities, organizational creativity and organization performance in public organizations. *Journal of Systems and Information Technology*. https://doi.org/10.1108/jsit-10-2022-0239	1	n.a.
Almheiri et al. (2026)	Leveraging strategic capabilities and institutional pressure for enhanced hotel resilience and performance. *Journal of Business and Socio-economic Development*. https://doi.org/10.1108/jbsed-02-2026-0134	1	n.a.
Alnofeli et al. (2025)	AI-powered CRM capability model: Advancing marketing ambidexterity, profitability and competitive performance. *International Journal of Information Management*. https://doi.org/10.1016/j.ijinfomgt.2025.102981	2	n.a.
Alnofeli et al. (2025)	Unlocking the power of AI in CRM: A comprehensive multidimensional exploration. *Journal of Innovation & Knowledge*. https://doi.org/10.1016/j.jik.2025.100731	1	n.a.
Alves et al. (2026)	Generative AI Adoption in B2B Firms: Ethical Governance, Innovation Capabilities, and Long-Term Competitive Performance. *Systems*. https://doi.org/10.3390/systems14040410	2	n.a.

Table C.1 continued

Study	Title, journal and identifier	Tier	Quality
Al-Mamary (2025)	The transformative power of artificial intelligence in entrepreneurship: exploring AI's capabilities for the success of entrepreneurial ventures. *Future Business Journal*. https://doi.org/10.1186/s43093-025-00533-7	2	n.a.
Al-Romeedy (2026)	AI-driven knowledge and intelligence: a strategic framework for competitive advantage in tourism and hospitality industry. *Business Process Management Journal*. https://doi.org/10.1108/bpmj-06-2025-1024	2	n.a.
Amin & Sivakumaran (2026)[T]	Harnessing artificial intelligence (AI) for innovation in family businesses: A conceptual model based on dynamic capabilities theory. *Strategic Business Research*. https://doi.org/10.1016/j.sbr.2026.100135	3	–
Anh et al. (2026)	The role of the level of artificial intelligence adoption on business model innovation, sustainable competitive advantage, and firm performance: Integrating the TOE framework and Dynamic Capabilities theory. *Green Technologies and Sustainability*. https://doi.org/10.1016/j.grets.2026.100384	2	n.a.
Anjum et al. (2026)[T]	From generative AI to sustainable performance: The mediating roles of knowledge management, intellectual capital, and competitive advantage in manufacturing SMEs. *Journal of Industrial Information Integration*. https://doi.org/10.1016/j.jii.2026.101125	2	n.a.
Arowolo (2026)	Artificial Intelligence and Strategic Management: A Theoretical Analysis of Technological Innovation, Competitive Advantage, and Sustained Organizational Performance in Businesses. *The International Journal of Interdisciplinary Cultural Studies*. https://doi.org/10.18848/rbk23y77	3	–
Awamleh & Bustami (2022)	Examine the Mediating Role of the Information Technology Capabilities on the Relationship Between Artificial Intelligence and Competitive Advantage During the COVID-19 Pandemic. *SAGE Open*. https://doi.org/10.1177/21582440221119478	2	n.a.
Ayoub & Sopuru (2026)	Reconfiguring Strategic Capabilities in the Digital Era: How AI-Enabled Dynamic Capability, Data-Driven Culture, and Organizational Learning Shape Firm Performance. *Sustainability*. https://doi.org/10.3390/su18031157	1	n.a.
Barabas et al. (2025)	Implementing A Company'S Competitive Strategy Using Artificial Intelligence. *Economic scope*. https://doi.org/10.30838/ep.196.144-151	3	–
Bidar et al. (2026)	Information systems strategy in the age of machine learning. *Business Horizons*. https://doi.org/10.1016/j.bushor.2026.07.001	3	–
Black et al. (2024)	Moving beyond 'proof points': Factors underpinning AI-enabled business model transformation. *International Journal of Information Management*. https://doi.org/10.1016/j.ijinfomgt.2024.102796	2	n.a.
Bunea & Corboș (2026)	When Does Artificial Intelligence Pay Off in Electronic Retailing? A Dual-Path Model from Implementation to Competitive Advantage. *Journal of theoretical and applied electronic commerce research*. https://doi.org/10.3390/jtaer21040119	2	n.a.
Callefi et al. (2026)	Unlocking competitive advantage with generative AI: a capability taxonomy for supply chains through the RBV lens. *Journal of Enterprise Information Management*. https://doi.org/10.1108/jeim-02-2026-0220	2	n.a.
Chen (2026)	AI-Augmented Management: Redefining Decision-Making, Leadership, and Organizational Performance. *Scholedge International Journal of Management & Development*. https://doi.org/10.19085/sijmd130201	3	–
Chen et al. (2026)	Why Sustainability-Oriented Resources Translate Differently Into Sustainable Performance and Durable Advantage: A Socio-Technical View of Green Innovation Ambidexterity. *Business Strategy and the Environment*. https://doi.org/10.1002/bse.71330	2	n.a.
Chowdhury et al. (2024)	Generative Artificial Intelligence in Business: Towards a Strategic Human Resource Management Framework. *British Journal of Management*. https://doi.org/10.1111/1467-8551.12824	3	–
Climent et al. (2024)	AI-enabled business models for competitive advantage. *Journal of Innovation & Knowledge*. https://doi.org/10.1016/j.jik.2024.100532	3	–
Csaszar et al. (2024)	Artificial Intelligence and Strategic Decision-Making: Evidence from Entrepreneurs and Investors. *Strategy Science*. https://doi.org/10.1287/stsc.2024.0190	3	–
Darwish et al. (2025)	Artificial intelligence's impact on theme parks' competitiveness: role of information systems and organisational agility. *Tourism Review*. https://doi.org/10.1108/tr-03-2024-0216	2	n.a.
Deng et al. (2025)	Generative AI-enabled intelligent auditing: an organizational adaptation mechanism study based on dynamic capability theory. *Future Technology*. https://doi.org/10.55670/fpll.futech.4.3.15	1	moderate
Al Dhaheri et al. (2024)	Do environmental turbulence, dynamic capabilities, and artificial intelligence force SMEs to be innovative?. *Journal of Innovation & Knowledge*. https://doi.org/10.1016/j.jik.2024.100528	1	n.a.
Dzreke (2025)	The Competitive Advantage of AI in Business: A Strategic Imperative. *International Journal For Multidisciplinary Research*. https://doi.org/10.36948/ijfmr.2025.v07i04.50400	2	low

Table C.1 continued

Study	Title, journal and identifier	Tier	Quality
Dzreke (2025)†	The symbiotic interplay between big data analytics (BDA) and artificial intelligence (AI) in the formulation and execution of sustainable competitive advantage: A multi-level analysis. *Frontiers in Research*. https://doi.org/10.71350/30624533119	2	moderate
Dzreke & Dzreke (2025)†	The algorithmic-based view: Why data, models, and systems replace VRIN as the core of competitive advantage. *Computer Science & IT Research Journal*. https://doi.org/10.51594/csitrj.v6i9.2061	3	–
Elshaer et al. (2025)	Artificial Intelligence-Driven Supply Chain Agility and Resilience: Pathways to Competitive Advantage in the Hotel Industry. *Logistics*. https://doi.org/10.3390/logistics10010005	2	n.a.
Falahat et al. (2026)	AI adoption and innovation in MSMEs for enhancing competitiveness and firm performance. *Benchmarking An International Journal*. https://doi.org/10.1108/bij-05-2026-0433	2	n.a.
De Fano et al. (2026)	Harnessing AI ambidexterity for competitive advantage: the role of dynamic capabilities in digital innovation ecosystems. *European Journal of Innovation Management*. https://doi.org/10.1108/ejim-11-2024-1404	1	n.a.
Feroz & Kwak (2024)	Digital Transformation (DT) and Artificial Intelligence (AI) Convergence in Organizations. *Journal of Computer Information Systems*. https://doi.org/10.1080/08874417.2024.2424372	2	n.a.
Firmansyah et al. (2025)†	Integration of Knowledge Management Capability (KMC) mediated by AI Capability and Organizational Learning Agility for Sustainable Competitive Advantage in the Digital Era in Binjai City MSMEs. *International Journal of Economic Technology and Social Sciences (Injects)*. OpenAlex W7217248548	2	low
Gafsi (2026)	Artificial intelligence capabilities, generational involvement, and competitive performance: Case of Tunisian family businesses. *Journal of Global Business Insights*. https://doi.org/10.5038/2640-6489.11.1.1358	2	n.a.
Gafsi (2026)	Digital Capabilities, Family Involvement, and Competitive Performance: A Moderated Mediation Perspective. *European Journal of Family Business*. https://doi.org/10.24310/ejfb.16.1.2026.21976	2	n.a.
Ghobakhloo et al. (2025)	Building antifragile manufacturing systems through strategic technology integration. *Journal of Manufacturing Technology Management*. https://doi.org/10.1108/jmtm-08-2025-0789	1	n.a.
Giang et al. (2026)	How Digital Transformation and Data-Driven Capabilities Foster Competitive Advantage: The Mediating Role of Organizational Agility in Vietnamese SMEs. *SAGE Open*. https://doi.org/10.1177/21582440261472113	2	n.a.
Guendouz (2024)	Artificial Intelligence-Powered Customer Experience Management (Moving from Mass to Hyper-Personalization in light of Relationship Marketing). *International Journal for Scientific Research*. https://doi.org/10.59992/ijsr.2024.v3n6p9	3	–
Guo et al. (2025)	Cross-border platforms' coupling of artificial intelligence adoption and strategic agility for upgraded ESMEs' global value chains: international marketing capability perspectives. *International Marketing Review*. https://doi.org/10.1108/imr-10-2024-0418	2	high
Haq et al. (2026)	Activating Resources for Sustainable Advantage: Green Innovation, AI Adoption and Institutional Support in Construction SMEs of an Emerging Economy. *Journal of Entrepreneurship and Innovation in Emerging Economies*. https://doi.org/10.1177/23939575261444792	2	n.a.
Hossain et al. (2022)†	Marketing analytics capability, artificial intelligence adoption, and firms' competitive advantage: Evidence from the manufacturing industry. *Industrial Marketing Management*. https://doi.org/10.1016/j.indmarman.2022.08.017	2	n.a.
Htet et al. (2026)†	Artificial intelligence (AI) and 21st-century sustainable marketing: A dynamic capabilities approach to sustainable competitive advantage. *Strategic Business Research*. https://doi.org/10.1016/j.sbr.2026.100166	1	n.a.
Huang et al. (2025)	Artificial Intelligence and Competitive Advantage of Tourism Enterprises: A Chain Mediation Effect. *Emerging Science Journal*. https://doi.org/10.28991/esj-2025-09-03-016	2	n.a.
Hussain et al. (2022)	Examining the role of artificial intelligence in determining sustainable competitive advantage: Evidence from the pharmaceutical sector of Karachi Pakistan. *Journal of Future Sustainability*. https://doi.org/10.5267/j.jfs.2022.11.003	2	n.a.
Immaculate & Njuguna (2026)	Ai Technology Integration And Competitive Advantage: Evidence From Telecommunications Firms In Nairobi City County, Kenya. *THE INTERNATIONAL JOURNAL OF MANAGEMENT SCIENCE AND BUSINESS ADMINISTRATION*. https://doi.org/10.5281/zenodo.20560605	2	n.a.
Jaruwanakul (2024)	The Influence of AI-CRM Adoption and Big Data Analytical Capability on Firm Performance of Large Enterprises in Thailand. *GLOBAL BUSINESS & FINANCE REVIEW*. https://doi.org/10.17549/gbfr.2024.29.2.112	2	low
Jum'a et al. (2026)	Harnessing big data and AI for competitive and sustainable performance: the mediating role of supply chain innovation in emerging economies. *Management & Sustainability An Arab Review*. https://doi.org/10.1108/msar-09-2025-0341	2	n.a.

Table C.1 continued

Study	Title, journal and identifier	Tier	Quality
Kadir et al. (2026)	Achieving Sustainable Competitive Advantage through Generative AI and Digital Leadership: The Mediating Role of Green Technology Innovation. *Bulletin of Management Review*. OpenAlex W7218010348	2	low
Kambara (2026)	Reflexive capabilities: Strategic adaptation in fashion's symbolic economy. *Fashion Style & Popular Culture*. https://doi.org/10.1386/fspc_00376_1	3	–
Karami (2026)	Reframing strategic decision-making as a dynamic capability: A conceptual framework for AI enabled organizations. *Advances in Business and Management Research*. https://doi.org/10.1016/j.abmr.2026.100005	3	–
Kemp (2024)	Competitive Advantage Through Artificial Intelligence: Toward a Theory of Situated AI. *Academy of Management Review*. https://doi.org/10.5465/amr.2020.0205	3	–
Kim & Seo (2026)	The Effect of Organizational IT Capabilities (AI Capabilities, Big Data Analytics Capabilities) on Competitive Advantage and Export Performance: The Moderating Role of Perceived External Export Stimuli. *Journal of the Korea Academia-Industrial cooperation Society*. https://doi.org/10.5762/kais.2026.27.9.125	2	n.a.
Kokash (2026)	Data Quality and AI Capabilities as Catalysts for Firm Performance in the Tourism and Recreation Industry. *Journal of Quality Assurance in Hospitality & Tourism*. https://doi.org/10.1080/1528008x.2026.2711976	1	n.a.
Kumar et al. (2023)	Automation-augmentation paradox in organizational artificial intelligence technology deployment capabilities; an empirical investigation for achieving simultaneous economic and social benefits. *Journal of Enterprise Information Management*. https://doi.org/10.1108/jeim-09-2022-0307	1	n.a.
Lai et al. (2026)	AIoT quality, supply chain intelligence, and sustainable competitive advantage: A study of Taiwan's semiconductor industry. *International Journal of Information Management Data Insights*. https://doi.org/10.1016/j.jjimei.2026.100453	2	n.a.
Le et al. (2025)	Optimizing and restructuring resources for sustainable firm performance in the AI era: the role of dynamic capabilities and circular manufacturing. *Sustainable Futures*. https://doi.org/10.1016/j.sftr.2025.101441	1	n.a.
Li & Sattayatham-rongthian (2026)[E]	Artificial intelligence orientation, ESG orientation, and opportunity sensing capability: The roles of absorptive capacity and entrepreneurial opportunity recognition in Yunnan's tourism industry. *Decision Science Letters*. https://doi.org/10.5267/j.dsl.2026.5.003	2	moderate
Li et al. (2025)[T]	Leveraging generative AI capabilities for competitive advantage: A moderated mediation analysis of environmental dynamism and service innovation. *Industrial Marketing Management*. https://doi.org/10.1016/j.indmarman.2025.05.007	2	n.a.
Li et al. (2026)	Do green innovation and resilience mediate the relationship between AI-enabled dynamic capabilities and sustainable supply chain performance?. *Cleaner Logistics and Supply Chain*. https://doi.org/10.1016/j.clscn.2026.100367	1	n.a.
Liu & ADIL (2026)	Digital Technologies and Sustainable Competitiveness: The Role of AI and Internet of Things in Driving Green Innovation and Firm Performance. *Corporate Social Responsibility and Environmental Management*. https://doi.org/10.1002/csr.70635	2	n.a.
Luo et al. (2025)[T]	ChatGPT's infusion advantage: The moderating role of strategic agility. *Journal of Strategy & Innovation*. https://doi.org/10.1016/j.jsinno.2025.200541	2	n.a.
Mahfoud & El-Adly (2026)	AI adoption and sustainable competitive advantage in SMEs: the roles of marketing analytics and innovation. *Journal of Entrepreneurship in Emerging Economies*. https://doi.org/10.1108/jeee-06-2025-0333	2	n.a.
Malik (2021)[T]	Beyond Digital Transformation: How Artificial Intelligence Reshapes Organizational Capabilities, Innovation, And Competitive Advantage. *TPM Testing, psychometrics, methodology in applied psychology*. https://doi.org/10.5281/zenodo.22806492	2	moderate
Mashi et al. (2024)	Enhancing Entrepreneurship Performance Through Artificial Intelligence Technology: Evidence from Manufacturing Firms in South-West, Nigeria. *International Journal of Advanced Studies in Economics and Public Sector Management*. https://doi.org/10.48028/iiprds/ijasepsm.v12.i2.14	2	low
Maune (2026)	Harnessing Competitive Intelligence and AI for Corporate Growth and Sustainability. *Journal of Intelligence Studies in Business*. https://doi.org/10.37380/jisib.v15i2.3105	1	low
Mili et al. (2025)	From code to quality: How AI is transforming quality management in Algerian startups. *International Journal of Innovative Research and Scientific Studies*. https://doi.org/10.53894/ijirss.v8i1.4802	2	n.a.
Mukhopadhyay et al. (2024)	Developing artificial intelligence enabled Marketing 4.0 framework: an Industry 4.0 perspective. *Qualitative Market Research An International Journal*. https://doi.org/10.1108/qmr-06-2023-0086	1	n.a.

Table C.1 continued

Study	Title, journal and identifier	Tier	Quality
Neira-Fernández et al. (2026)	The impact of artificial intelligence and facilitating conditions on digital maturity: the mediating role of digital dynamic capabilities. *Management Research The Journal of the Iberoamerican Academy of Management*. https://doi.org/10.1108/mrjiam-06-2025-1748	2	n.a.
Nguyen et al. (2025)	Artificial intelligence in accounting, risk management, sustainable competitiveness and managerial IT infrastructure: a moderation-mediation model. *Asia-Pacific Journal of Business Administration*. https://doi.org/10.1108/apjba-02-2025-0101	2	n.a.
Nikzat & Noorymotlagh (2025)	Artificial Intelligence in Business: Driving Innovation and Competitive Advantage. *International journal of industrial engineering and operational research*. OpenAlex W7215671917	2	n.a.
Noor et al. (2026)†	Generative Ai And Entrepreneurial Marketing Performance In E-Commerce Firms. *Perspectives in Social Sciences*. OpenAlex W7221738716	1	moderate
Pesqueira (2025)	The strategic role of blockchain-enabled federated learning, AI agents and dynamic capabilities in driving pharmaceutical operational advantages. *Journal of AI, robotics & workplace automation.*. https://doi.org/10.69554/ufhd8035	1	n.a.
Nguyen & Nguyen (2026)	Entrepreneurial orientation, dynamic capabilities, and startup performance: The amplifying role of AI. *Problems and Perspectives in Management*. https://doi.org/10.21511/ppm.24(2).2026.49	1	moderate
Ramírez et al. (2026)[T]	Artificial intelligence and organisational performance in the tourism sector: An integrative model linking enterprise architecture and dynamic capabilities theory. *Social Sciences & Humanities Open*. https://doi.org/10.1016/j.ssaho.2026.103264	1	n.a.
Romano & Conti (2026)	An Integrated Model for Data-Driven Decision-Making and AI-Enabled Optimization to Enhance Long-Term Competitive Performance in Enterprises. *International Journal of Emerging Innovations in Science and Technology*. OpenAlex W7220221707	3	–
Ruangkanjanases et al. (2025)	Achieving green competitive advantage through generative AI: the mediating roles of organisational creativity and green innovation ambidexterity in manufacturing. *International Journal of Logistics Research and Applications*. https://doi.org/10.1080/13675567.2025.2573664	2	n.a.
Safwat et al. (2026)	How Artificial Intelligence Shapes Sustainable Competitive Advantage through Market Knowledge in Tourism and Hospitality Organizations. *International Journal of Tourism and Hospitality Management*. https://doi.org/10.21608/ijthm.2026.522480	2	n.a.
Samadhiya et al. (2023)[T]	Artificial intelligence - partner relationships management for climate management in B2B firms to achieve sustainable competitiveness. *Industrial Marketing Management*. https://doi.org/10.1016/j.indmarman.2023.11.002	2	n.a.
Sandeep & Lavanya (2025)	Artificial intelligence redefines the source of competitive advantage in recruitment and selection: a resource-based view and dynamic managerial capabilities approach. *Journal of Management Development*. https://doi.org/10.1108/jmd-03-2024-0082	2	n.a.
Sandeep et al. (2025)	Leveraging AI in recruitment: enhancing intellectual capital through resource-based view and dynamic capability framework. *Journal of Intellectual Capital*. https://doi.org/10.1108/jic-05-2024-0155	1	n.a.
Shao et al. (2026)	The influence of AI capability on enterprise competitive advantage: the mediating effect of business model innovation. *Journal of Enterprise Information Management*. https://doi.org/10.1108/jeim-08-2024-0453	2	n.a.
Sharma (2026)	AI-Augmented marketing decision-making and competitive performance: A resource-based view of capability orchestration. *Acta Psychologica*. https://doi.org/10.1016/j.actpsy.2026.107351	2	n.a.
Shonubi (2026)	Unwrapping generative AI paradigms for product and service innovation differentiation. *Journal of Strategy & Innovation*. https://doi.org/10.1016/j.jsinno.2025.200562	3	–
Shore et al. (2024)[E]	Building entrepreneurial resilience during crisis using generative AI: An empirical study on SMEs. *Technovation*. https://doi.org/10.1016/j.technovation.2024.103063	2	n.a.
Siaw & Ali (2025)	Substitution and complementarity between human and artificial intelligence: a dynamic capabilities view. *Journal of Managerial Psychology*. https://doi.org/10.1108/jmp-06-2024-0398	3	–
Singh et al. (2025)	Exploring the role of artificial intelligence on business innovation and entrepreneurship. *Strategy and Leadership*. https://doi.org/10.1108/sl-04-2025-0080	2	n.a.
Sofiyan et al. (2026)[E]	Transforming naval organizations in the digital age: leadership, AI and dynamic capabilities for change. *Management Decision*. https://doi.org/10.1108/md-10-2025-2947	2	n.a.
Son et al. (2024)	Effects of AI Marketing (AIM) of Small Businesses on Competitive Advantage Performance and Continuous Use Intention by Mediating Absorption Capabilities. *The Journal of Next-generation Convergence Technology Association*. https://doi.org/10.33097/jncta.2024.08.10.2310	2	n.a.

Table C.1 continued

Study	Title, journal and identifier	Tier	Quality
Srahin et al. (2026)	An empirical study of augmented analytics adoption and its impact on firms' agility through AI-based models. *Frontiers in Artificial Intelligence*. https://doi.org/10.3389/frai.2026.1940124	2	moderate
STOICA & SOCOL (2025)	The Development Of Competitive Capabilities In Modern Organizations Through Artificial Intelligence Algorithms. *BUSINESS EXCELLENCE AND MANAGEMENT*. https://doi.org/10.24818/beman/2025.s.i.5-17	3	–
Suleiman et al. (2024)	Artificial intelligence: a catalyst to organizational learning and development in 21st century. *Development in Learning Organizations An International Journal*. https://doi.org/10.1108/dlo-02-2024-0037	2	n.a.
Tavares & Brandão (2025)	The Interplay Of Corporate Governance Mechanisms And Digital Transformation On Firm Performance And Competitive Advantage. *Review of Accounting, Finance and Economics*. OpenAlex W7220149329	2	low
Terrell & Woods (2026)	Organizational AI Acculturation: How Human- AI Interaction Encodes Institutional Knowledge and Builds Competitive Advantage. *Knowledge and Process Management*. https://doi.org/10.1002/kpm.70155	3	–
Texeira (2026)	From AI resources to sustainable competitive advantage: Building AI-enabled sustainable gastronomy capabilities in restaurants. *Journal of Tourism Theory and Research*. https://doi.org/10.24288/jttr.2012042	2	high
Tian et al. (2023)[E]	Enhancing innovativeness and performance of the manufacturing supply chain through datafication: The role of resilience. *Computers & Industrial Engineering*. https://doi.org/10.1016/j.cie.2023.109841	2	n.a.
Tio et al. (2026)	Beyond Automation: A Conceptual Framework for Strategic AI Integration in Business Decision-Making Processes. *ADPEBI International Journal of Business and Social Science*. https://doi.org/10.54099/aijbs.v6i2.1951	3	–
Tu et al. (2026)	Can AI Capability Boost Firm Competitiveness? A Serial Mediation Analysis Based on Organizational Learning and Organizational Resilience. *Systems*. https://doi.org/10.3390/systems14060667	2	n.a.
TYTARENKO (2025)	Artificial intelligence as a factor in optimising strategic management and strengthening the competitive position of an enterprise. *Economics Finances Law*. https://doi.org/10.37634/efp.2025.2.6	3	–
Wahyudi et al. (2025)	Green product innovation, R&D, and AI adoption: The moderating role of intellectual capital in achieving competitive advantage. *Journal of Accounting and Investment*. OpenAlex W7220587514	2	n.a.
Wang et al. (2026)	Pathways to sustainable competitive performance: social entrepreneurship orientation, disruptive innovation and artificial intelligence capabilities. *Humanities and Social Sciences Communications*. https://doi.org/10.1057/s41599-026-06851-7	2	high
Waqas et al. (2024)	Technological management capabilities as a pathway toward green production and green competitive advantage. *Journal of Manufacturing Technology Management*. https://doi.org/10.1108/jmtm-10-2023-0456	2	n.a.
Wasesa et al. (2025)	Integration of Artificial Intelligence in Management Information Systems to Improve the Effectiveness of Strategic Decision-Making in the Digital Era. *Journal of Computer Science Information Technology and Telecommunication Engineering*. https://doi.org/10.30596/jcositte.v6i2.26051	2	low
Wei et al. (2025)	Algorithmic Dynamic Capabilities: Orchestrating Ecosystem Strategy and Competitive Advantage in the Age of Artificial Intelligence. *Frontiers in Business Innovations and Management*. OpenAlex W7218036322	3	–
Wenting et al. (2026)	AI Adoption in Decision-Making and Firm Competitiveness in Tourism: Integrating CTUAT with Resource-Based and Dynamic Capability Perspectives. *Journal of Quality Assurance in Hospitality & Tourism*. https://doi.org/10.1080/1528008x.2026.2639381	2	n.a.
Wu et al. (2026)[T]	How AI matters for competitive advantage: a managerial cognition perspective on managing the triple role of AI and dynamic capabilities. *Journal of Business Research*. https://doi.org/10.1016/j.jbusres.2026.116413	1	n.a.
Yang et al. (2026)	Pathway to achieve sustainable value through task-AI fit: the critical roles of green knowledge management capabilities, competitive advantage and financial resilience. *Journal of Knowledge Management*. https://doi.org/10.1108/jkm-03-2026-0431	2	n.a.
Yoo et al. (2024)	The impact of AI-enabled CRM systems on organizational competitive advantage: A mixed-method approach using BERTopic and PLS-SEM. *Heliyon*. https://doi.org/10.1016/j.heliyon.2024.e36392	2	n.a.
Yue (2026)	Rethinking Competitive Strategy for a World Driven by Intelligent Technologies. *Scholedge International Journal of Business Policy & Governance*. https://doi.org/10.19085/sijbpg130203	3	–
Zhang (2026)	Ai-infused business model innovation for competitive advantage in the Era of big data and digital transformation. *Egyptian Informatics Journal*. https://doi.org/10.1016/j.eij.2026.101046	2	n.a.

Table C.1 continued

Study	Title, journal and identifier	Tier	Quality
Zhou (2025)	AI Empowerment and Measurement: Reconstructing the Enterprise Digital Maturity Model and its Implications for Service Ecosystem. *Frontiers in Sustainable Development*. https://doi.org/10.54691/tt5b9s51	3	–
Zhu (2025)	Enhancing Competitive Advantage through AI and Digital Technology: A Case Study of Jiangling Motors Group. *Asia Pacific Economic and Management Review*. https://doi.org/10.62177/apemr.v2i1.150	2	low

Source: author's own work, based on the review data.

Declaration of Honor

I hereby declare that I have written this thesis independently and without outside help, that I have used no sources or aids other than those indicated, and that I have marked all passages taken literally or in substance from other works as such. This thesis has not been submitted in the same or a similar form to any other examination authority.

The aids used include a large language model (Claude, Anthropic), which supported the screening of records, the extraction of data and its verification as described in Sections 3.2.2 and 3.4.

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Place, Date

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Vishalini Kalsan James Alias Kannan